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EXPERIMENTAL ANALYSIS AND COMPENSATION OF MOTOR – INDUCED MAGNETIC INTERFERENCE IN UAVs USING EMBEDDED SYSTEM

A PROJECT REPORT

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ABSTRACT

This project proposes low-cost, lightweight embedded system designed to characterize and compensate the magnetic field interference that is induced from the Unmanned Aerial Vehicles (UAVs). According to the research magnetic interference generates and induced by high current propulsion systems and power distribution systems of the UAVs significantly degrades magnetometer measurements which are essential for geophysical exploration, reducing data reliability, which frequently masks subtle the magnetic signals. The system utilizes digital compensation using a micro controller integrated with sensors calibrated with brushless motor which quantifies the throttle percentage, measures current draw and magnetic deviation. A mathematical model is introduced for the compensation function. This system employs an active data logging approach by recording real-time motor current consumption synchronized with magnetic heading data, a mathematical correlation between propulsion load and magnetic deviation can be established. The results show that by monitoring current fluctuations, a digital compensation algorithm can effectively compensate and bring the system linear using survey data. This study presents a scalable framework for increasing the signal-to-noise ratio in small-scale aeromagnetic surveys, thereby improving the detectability of geomagnetic anomalies.

Keywords: UAV magnetometry, electromagnetic interference, correlation, data compensation.

INDEX

s.no	Content	Page no
01.	Introduction	7
02.	Problem Statement	8
03.	Objectives	9
04.	Project Outcome	10
05.	Applications	11
06.	Literature Survey	12
07.	Methodology	21
08.	Electromagnetic Interference (EMI) And Compensation Techniques	24
09.	Components Selection	30
10.	Circuit design	35
11.	System Architecture	39
12.	Experimental Section	45
13.	Results And Discussion	59
14.	Conclusion	70
15.	Reference	73

ABBREVIATION

1. ESC	Electronic speed controller
2. UAV	Unmanned Aerial Vehicle
3. IMU	Inertial Measurement Unit
4. ADC	Analog to digital converter
5. I ² C	Inter-Integrated Circuit
6. PWM	Pulse Width Modulation
7. B	magnetic flux density
8. B _{raw}	unfiltered magnetic data
9. B _{measured}	measured magnetic field
10. B _{offset}	Offset Magnetic Field
11. K _x ,K _y ,K _z	Linear Compensation Coefficients
12. K ₂	Non-linear Compensation Coefficient
13. r	Offset Corrected Magnetic Field
14. c	Compensated Magnetic Field
15. B _{comp}	compensated (clean) magnetic data
16. I	electrical current
17. V _{out}	sensor output voltage
18. V _{ref}	reference voltage (calibration baseline)
19. K	magnetic compensation coefficient
20. F _s	sampling frequency
21. μ	magnetic permeability
22. T	time
23. A _L	loop area
24. V _{ind}	induced voltage
25. H	magnetic field strength
26. Z(ω)	impedance (frequency dependent)
27. Φ_B	magnetic flux
28. μ_o	permeability of free space
29. μ_r	relative permeability
30. α	SmoothingFactor

1.INTRODUCTION

Unmanned Aerial Vehicles (UAVs) have revolutionized geophysical exploration by enabling low-cost, high-resolution aeromagnetic surveys over challenging terrains. However, brushless motor propulsion systems and power electronics in small UAVs generate significant electromagnetic interference that corrupts magnetometer measurements—critical for heading determination and geomagnetic anomaly detection. High-current drawing through Electronic Speed Controllers (ESCs) induces magnetic field distortions that mask subtle geophysical signals, compromising survey accuracy and data reliability.

This project presents a low-cost, lightweight embedded compensation system designed to characterize and compensate motor-induced magnetic interference in UAV magnetometry. The proposed architecture integrates an Arduino Nano microcontroller with a calibrated sensor suite—QMC5883L triaxial magnetometer (± 1.3 Gauss), ACS712-30A current sensor (66 mV/A sensitivity), and A2212/1000KV brushless motor interfaced through SimonK 30A ESC. Custom C++ firmware enables synchronized real-time data acquisition (100 Hz) of throttle position, instantaneous current consumption, and 3-axis magnetic field deviations.

The system implements a comprehensive dataflow pipeline: baseline calibration of all components, linear compensation, and non-linear correction if the system is non-linear, and exponential moving average smoothing. Mathematical modeling establishes direct correlation between propulsion current and magnetic deviation, enabling digital compensation that restores linear magnetometer response independent of motor load. Results demonstrate effective interference suppression, achieving magnetic field compensation suitable for small-scale aeromagnetic surveys.

This scalable framework significantly enhances signal ratio in UAV-based geomagnetic exploration, providing a practical solution for interference compensation without additional hardware mass or cost. The system's active data logging and compensation algorithms offer a foundation for reliable geophysical data collection from resource-constrained aerial platforms.

2. PROBLEM STATEMENT

In Unmanned Aerial Vehicles (UAVs), the magnetometer is an important sensor used for heading estimation and navigation. However, during motor operation, the electric current flowing through the motor and Electronic Speed Controller (ESC) generates magnetic disturbances. These disturbances affect the magnetometer readings and reduce the accuracy of orientation and navigation. As a result, the UAV may experience incorrect heading estimation and unstable flight performance. Therefore, it is necessary to experimentally analyse the motor-induced magnetic interference and develop an embedded system-based compensation method to improve the reliability and accuracy of magnetometer measurements in UAV systems.

3. OBJECTIVES

- To analyze the effect of motor-generated magnetic disturbances on the magnetometer in UAV systems.
- To measure the magnetic field variations caused by the motor during operation using sensors and embedded hardware.
- To develop an embedded system-based method to compensate the motor-induced magnetic interference.
- To improve the accuracy and reliability of magnetometer readings in UAV navigation.
- To validate the compensation technique through experimental testing and data analysis.

4. PROJECT OUTCOME

- Successful experimental analysis of motor-induced magnetic interference in UAV systems.
- Development of an embedded system to monitor motor current and magnetometer readings.
- Implementation of a compensation technique to reduce the effect of motor-generated magnetic disturbances.
- Improved accuracy and stability of magnetometer data during motor operation.
- A reliable method that can enhance UAV navigation and heading estimation performance.

5. APPLICATIONS

- UAV Navigation Systems – Improves heading accuracy of magnetometers used in Unmanned Aerial Vehicles for stable flight and precise direction control.
- Drone Mapping and Surveying – Enhances orientation accuracy for drones used in land surveying, mapping, and environmental monitoring.
- Autonomous Drone Systems – Supports reliable sensor data for autonomous flight control and navigation algorithms.
- Defence and Surveillance – Helps military and security drones maintain accurate directional information during missions.
- Robotics and Embedded Systems – The interference compensation method can be applied to robotic platforms that use magnetometers near electric motors.
- Research and Development – Useful for improving sensor performance in experimental UAV and embedded system research.

6. LITERATURE SURVEY

6.1 INTRODUCTION

Unmanned Aerial Vehicles (UAVs) have become an important platform for various sensing and monitoring applications due to their flexibility, low operational cost, and ability to access difficult or hazardous environments. In recent years, UAV-based magnetic sensing systems have gained significant attention in fields such as mineral exploration, geological mapping, infrastructure inspection, unexploded ordnance detection, environmental monitoring, archaeological surveys, and defence surveillance. By integrating magnetometers with UAV platforms, it is possible to measure variations in the Earth's magnetic field and detect buried objects, underground structures, and magnetic anomalies. However, one of the major challenges in UAV-based magnetic surveys is the magnetic interference generated by the aircraft itself. Components such as electric motors, power cables, servos, and onboard electronics produce electromagnetic fields that can distort the magnetometer readings and reduce measurement accuracy. To overcome this issue, several researchers have proposed different approaches including hardware design improvements, optimized sensor placement, magnetic interference testing, and software-based compensation techniques. The following literature survey reviews significant research works related to UAV magnetic interference, aeromagnetic compensation methods, and the development of UAV-based magnetic sensing systems.

6.2 REVIEW OF EXISTING WORKS

Section 1 - Background: UAV Magnetic Surveys and Platform Reviews

6.2.1 Yaoxin Zheng et al. (2021)

UAV Platform Selection and Interference Suppression Review

Zheng et al. (2021) reviewed 70 studies on UAVs for magnetic surveys, comparing four platform types — fixed-wing, helicopter, multi-rotor, and airship — across applications like mineral exploration, volcano monitoring, and landmine detection. Multi-rotor UAVs emerged as the most popular choice due to their low cost, ease of use, and ability to fly at very low altitudes, delivering data quality comparable to traditional ground surveys at nearly half the cost (USD 35/km vs USD 85/km). The core challenge identified was magnetic interference from UAV motors and electronics, which can corrupt sensor readings by up to 800 nT. Two suppression approaches were reviewed: physically distancing the magnetometer from the

drone and applying mathematical compensation algorithms. However, the standard Tolles-Lawson compensation model used in manned aircraft cannot be directly applied to UAVs due to low flight altitudes and inability to perform required calibration maneuvers. The authors concluded that while UAV magnetic surveys are promising and cost-effective, interference suppression — especially for multi-rotor drones — remains an unsolved research problem

6.2.2 Kaub et al. (2021)

UAS Software for Comparing Sensor Systems & Compensation Methods

Kaub and colleagues studied how to reduce magnetic interference in UAV aeromagnetic surveys. They compared a tethered magnetometer setup with a software compensation method that removes aircraft-generated magnetic noise. The tethered approach reduced interference, but the long suspension made the sensor less stable during flight. The compensated fixed-mount setup gave better overall stability and more reliable survey results after processing. Their study shows that sensor placement and compensation are both important for accurate drone magnetic surveys.

6.2.3 Yong-Zai Xi et al. (2023)

Fixed-Wing UAV vs Unmanned Helicopter Aeromagnetic System

This study says the UAV itself creates magnetic interference that can affect aeromagnetic measurements. It compares fixed-wing and helicopter platforms and shows that the noise comes from both ferromagnetic parts and electric currents inside the aircraft. The paper mainly measures and maps that interference rather than giving a full compensation algorithm. Its result is that the interference can be strong enough to distort survey data, so platform design and sensor placement matter a lot. In short, it is a characterization study about magnetic disturbance, not a complete mathematical correction model.

Section 2 – Hardware Sensor Systems & System Prototyping

6.2.4 Sierra Luoma & Xiaobing Zhou

Construction of a Fluxgate Magnetic Gradiometer for Integration with a UAS

Luoma and Zhou (2020) designed a lightweight fluxgate magnetic gradiometer for use with a drone to survey underground magnetic features. The system used two fluxgate magnetometers and an Arduino Mega microcontroller for data collection and storage. It was

towed two meters below a DJI Matrice 600 hexacopter to reduce magnetic noise from the drone. Field tests in Montana showed the system could successfully collect magnetic and GPS data during flight. Results confirmed the system's feasibility for UAV-based magnetic surveying, with minor design improvements recommended for better accuracy.

6.2.5 Antonino Pisciotta et al. (2021)

A Lightweight Prototype of a Magneto metric System for UAV

This paper presents a lightweight magneto metric system mounted on a UAV for measuring Earth's magnetic anomalies. The authors built a compact setup with a 3-axis fluxgate magnetometer, GPS, data storage, and radio transmission. They tested it in the lab and at an archaeological site to see whether it could detect real magnetic anomalies. The system worked reliably and was able to identify low-amplitude anomalies linked to buried archaeological structures. The paper mainly focuses on hardware design and field testing, not on a detailed magnetic compensation model. It mentions magnetic interference and reduces it mainly by placing the sensor away from the drone, rather than by advanced mathematical correction

6.2.6 Nan Zhang et al. (2025)

UAV Aeromagnetic System Based on Fluxgate Sensor & Applications

Zhang and colleagues developed a UAV-based aeromagnetic survey system using a fluxgate magnetometer. They tested the system in field conditions to measure magnetic signals from geological targets. The study showed that the UAV platform can collect usable magnetic data for survey purposes. Their results suggest the system has potential for mineral exploration and magnetic anomaly detection. Overall, the research demonstrates that UAV-based magnetic surveying is a practical and promising approach.

6.2.7 Ning Lu et al. (2023)

Hybrid Fixed-Wing UAV Aeromagnetic Survey

This paper presents a hybrid fixed-wing UAV system for aeromagnetic surveying, designed to collect magnetic data more efficiently than traditional methods. It also studies magnetic interference from the UAV itself and uses a compensation model to reduce that noise. The field test at the Chating deposit showed that the system could collect usable low-altitude magnetic data. After processing, the magnetic anomalies helped identify geological structures

related to mineralization. So, the main result is that the UAV platform worked well for both magnetic data acquisition and geological interpretation.

Section 3 – Core Problem: Characterizing & Modelling Magnetic Interference

6.2.8 Bingyang Chen et al. (2023)

Magnetic Interference Analysis & Compensation of Airborne Electronics in UAV

Chen and colleagues analyzed magnetic interference generated by airborne electronic equipment in UAVs and found that the traditional Tolles–Lawson model was ineffective for UAV platforms. The study demonstrated that interference mainly depends on the power supply current of onboard systems. A current-based linear compensation model was proposed, reducing the standard deviation of interference from 15.56 nT to 1.52 nT, achieving an improvement ratio greater than 10.

6.2.9 Guocan Cao et al. (2020)

Real-Time Magnetometer Calibration via RLS/ML

This paper proposes a real-time calibration method for three-axis magnetometers using RLS and ML. It does not try to remove magnetic interference itself; instead, it builds a mathematical error model and corrects the sensor readings. The authors show that magnetometers are affected by ambient magnetic disturbances, so calibration is necessary before use. Their method estimates calibration parameters continuously while the sensor is running. The main result is that the proposed method gives better accuracy and faster computation than the traditional batch least-squares method. The paper also says the method can work in different magnetic-disturbance environments and may be used for other three-axis sensors too.

6.2.10 Zhenning SU et al. (2022)

Aeromagnetic Compensation Method Based on Ridge Regression Algorithm

Su et al. (2022) proposed an improved aeromagnetic compensation technique based on ridge regression. Traditional least-squares methods often face instability when variables are highly correlated due to multicollinearity in the Tolles-Lawson equation. By introducing a regularization term, ridge regression improves the stability of compensation parameters. Experimental results, evaluated using standard deviation as the performance metric, showed

that this method effectively reduced magnetic interference compared to conventional approaches.

Section 4 – Compensation Methods: Traditional to Advanced

6.2.11 You li et al. (2022)

Magnetic Interference Compensation Without Current Sensors

Li et al. proposed a magnetic interference compensation method based on a Multi-Source Two-channel Linear Time-Invariant (MTLI) model, which eliminates the need for additional current sensors. The method utilizes two magnetometers along with a multi-source correlation approach to estimate and remove interference. Experimental results demonstrated that the approach effectively reduces magnetic noise under both steady and maneuvering flight conditions. The study shows that magnetic interference can be accurately modeled and compensated using only magnetic measurements, resulting in a lighter and more efficient system suitable for UAV applications

6.2.12 Liwei Ye et al. (2024)

An Aeromagnetic Compensation Strategy for Large UAVs

Ye et al. (2024) improved aeromagnetic compensation for large UAVs by expanding the Tolles–Lawson model to 36 coefficients using INS/GPS data alongside a fluxgate magnetometer. They introduced a new algorithm, LRNM, which reduced magnetic noise from 2.211 nT to 0.121 nT. Compared to traditional methods, LRNM achieved 34% and 27% better performance than Least Squares and SVD methods respectively.

6.2.13 Zizhou Chen et al. (2025)

Intelligent Dynamic-Enhanced Compensation for UAV Magnetic Interference

This paper studies magnetic interference compensation for UAVs, not a new magnetic field itself. It improves the traditional Tolles–Lawson compensation model by adding flight attitude and angular-rate features. These added terms help describe the UAV’s self-generated magnetic interference more accurately. Then the authors use a neural-network-based method to predict and reduce interference. Their results show the new model compensates for

magnetic noise better than the original method. So, the paper is mainly about mathematical and intelligent compensation methods for cleaner UAV magnetic measurements.

Section 5 – AI/ML – Based Compensation & comparative validation

6.2.14 Ping Yu et al. (2025)

Quadrotor UAV Compensation via Time-Frequency Neural Network

Yu and colleagues proposed a compensation method combining continuous wavelet transform with a bi-directional short-term memory neural network. This method extracts time-frequency features from magnetic signals and automatically learns interference patterns generated by UAV motors. Experimental results showed improved performance compared with traditional filtering and regression methods, particularly under dynamic flight conditions

6.2.15 Weiming Dai et al. (2025)

Aeromagnetic Compensation for UAVs Using Transformer Neural Networks

This research presents a Transformer-based method for aeromagnetic compensation in UAVs. The main problem is that UAV magnetic measurements are disturbed by the aircraft's own magnetic interference, which reduces the accuracy of aeromagnetic surveys. To solve this, the authors combined the Tolles–Lawson model with a Transformer neural network to learn and remove the interference from the measured magnetic data. They tested the method on both simulated and real flight data and compared it with a traditional multilayer perceptron (MLP). The results showed that the Transformer model achieved better fitting ability and higher compensation accuracy, making it more effective for correcting magnetic interference in UAV aeromagnetic surveys

6.3 SUMMARY OF LITERATURE

Table 6.1 summary of literature review

S.NO	Title	Author / year	Method	Conclusion / key finding
1	UAVs for Magnetic Surveys: A Review on Platform Selection and Interference Suppression	Zheng et al., 2021	Systematic review of 70 studies. Classifies UAV types and interference sources (permanent, induced, eddy-current). Reviews suppression via tethering and Tolles-Lawson compensation.	Multi-rotor UAVs are most widely used. Sensor-UAV distance is the common suppression method but causes positional errors. Robust compensation

				models needed for body-mounted sensors.
2.	Quadrotor UAV Aeromagnetic Compensation via Time-Frequency Joint Representation Neural Network for Mineral Exploration	Liu et al., 2025	Time-Frequency Joint Representation CNN converts motor current and attitude signals into frequency images to predict and subtract magnetic interference.	TFJRNN reduces interference residuals >60% over Tolles-Lawson. Current sensor signals carry rich frequency-domain features useful for neural compensation.
3	An Aeromagnetic Compensation Strategy for Large UAVs	Wang et al., 2024	Extends 16-coefficient Tolles-Lawson model for large UAVs. Uses fluxgate reference magnetometer and ridge regression for coefficient estimation.	Extended TL model reduces heading error significantly. Calibration maneuver design (figure-of-eight) is critical for accuracy.
4	Magnetic Interference Analysis and Compensation Method of Airborne Electronics in a UAV	Yang et al., 2023	Measures interference from motors, ESCs, and power cables individually. Develops a current-correlated superposition compensation model.	Motors and cables are dominant interference sources, scaling with current draw. Current-correlated linear model reduces field error below 5 nT in steady state.
5	Aeromagnetic Compensation Method Based on Ridge Regression Algorithm	Pang et al., 2022	Applies L2 ridge regression to Tolles-Lawson model to solve multicollinearity in limited calibration data. Compared with OLS and PCA regression.	Ridge regression improves coefficient stability and reduces overfitting — practical for low-cost UAVs with limited calibration flights.
6	Construction of a Fluxgate Magnetic Gradiometer for Integration with an Unmanned Aircraft System	Cunningham et al., 2020	Designs and builds a dual-sensor fluxgate gradiometer for UAS. Tests vibration isolation, shielding, and validates against ground surveys.	Properly shielded fluxgate gradiometers achieve sub-nT resolution. Motor vibration isolation and ESC shielding are critical design requirements.
7	A Magnetic Interference Compensation Method for Airborne Electronic Equipment Without Current Sensors	Di et al., 2022	Uses only magnetometer output and IMU attitude data. SVD of directional cosine matrix estimates interference coefficients — no current sensor needed.	Current sensors are not strictly necessary under steady flight. SVD method degrades during rapid maneuvers, highlighting value of current-sensor-assisted compensation.
8	Aeromagnetic Compensation for UAVs Using Transformer Neural Networks	Chen et al., 2025	Transformer model with multi-head self-attention learns temporal dependencies among current, attitude, and speed variables to predict interference.	Transformers outperform LSTM/RNN for temporal interference modeling. Attention mechanism weights high-interference flight phases more heavily.
9	UAV Aeromagnetic System Based on Fluxgate Sensor and Its Applications	Zhang et al., 2025	Develops a full UAV survey system with 3-component fluxgate sensor, onboard noise suppression, and field validation against ground	Fluxgate UAV systems achieve survey-grade accuracy with proper compensation. System

			truth.	integration (power isolation, vibration damping) is as important as the algorithm.
10	Magnetic Surveys With UAS: Software for Comparing Sensor Systems, Suspension Designs & Compensation Methods	Kaub et al., 2021	Open-source Python software compares scalar (Cs-vapor) and vector (fluxgate) systems with tethered vs. fixed-mount configs over repeated grid flights.	Fixed-mount vector systems with compensation match tethered scalar systems. Open tools are essential for reproducible UAV magnetic survey comparison.
11	A Lightweight Prototype of a Magnetometric System for UAVs	Pisciotta et al., 2021	Low-cost COTS magnetometer system (<300g) integrated on a small UAV. Evaluates sensor noise, UAV electromagnetic interference, and field performance.	Sub-nT resolution is achievable with low-cost COTS sensors at adequate motor separation. Demonstrates feasibility without expensive survey-grade instruments.
12	Real-Time Calibration of Magnetometers Using the RLS/ML Algorithm	Vevelak et al., 2020	Recursive least squares / maximum likelihood algorithm for real-time hard-iron and soft-iron calibration during flight, without dedicated maneuvers.	Real-time RLS/ML outperforms static pre-flight calibration. Compensates time-varying hard-iron effects from temperature change and battery discharge.
13	Intelligent Dynamic-Enhanced Compensation for UAV Magnetic Interference	Li et al., 2025	Combines Tolles-Lawson baseline with a GRU neural network to capture nonlinear, time-varying residual interference not covered by TL model.	Hybrid physics+ML approach outperforms either method alone. GRU corrects dynamic residuals during aggressive maneuvers that TL cannot model.
14	Development of a Hybrid Fixed-Wing UAV Aeromagnetic Survey System (Chating Deposit Study)	Xu et al., 2023	Hybrid VTOL UAV deployed for mineral survey with TL compensation, diurnal drift correction, and leveling. Compared against ground magnetic data.	Hybrid VTOL UAVs combine fixed-wing coverage with VTOL flexibility. Compensated data agreed closely with ground truth, validating field deployment.
15	Study on the Aeromagnetic System: Fixed-Wing UAV vs. Unmanned Helicopter	Mu et al., 2023	Identical sensor payloads on fixed-wing UAV and helicopter over same area. Evaluates noise, heading error, compensation effectiveness, and anomaly detection.	Fixed-wing UAVs produce lower noise due to greater stability; helicopters adapt better to complex terrain. Interference characteristics differ per platform, requiring platform-specific compensation.

6.4 RESEARCH GAP

The reviewed studies collectively advance UAV aeromagnetic compensation but leave the following gaps unaddressed:

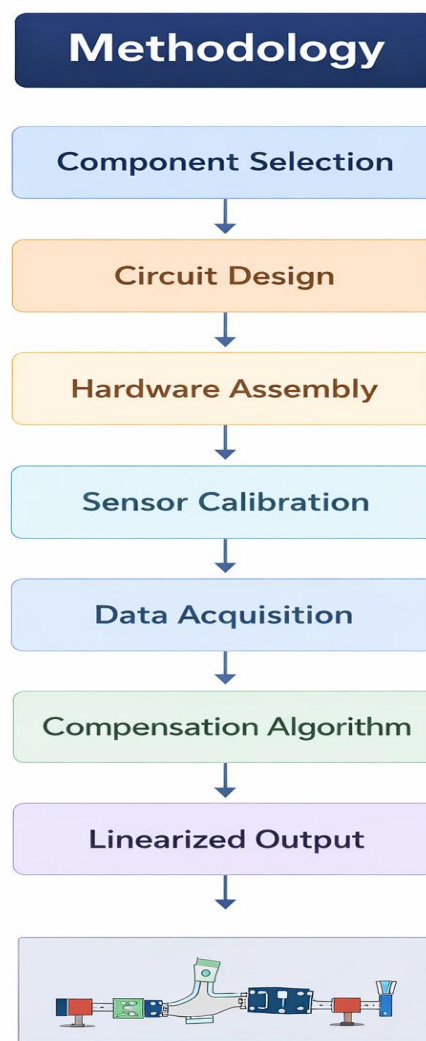
- No study uses a low-cost microcontroller (Arduino Nano) with a commercial Hall-effect current sensor for real-time magnetic interference compensation on a small drone.
- Di et al. (2022) argue compensation works without current sensors, while Yang et al. (2023) show current is the dominant interference source — no study directly compares both approaches on the same small multi-rotor platform.
- No study systematically compares a lab-assembled compensated magnetometer system against simultaneously acquired data from a real commercial drone's built-in magnetometer under identical motor load conditions.
- All existing compensation algorithms (TL regression, ridge regression, neural networks) run offline in post-processing. No embedded real-time compensation pipeline on a resource-constrained MCU has been demonstrated.
- This project addresses these gaps by building an Arduino Nano + ACS712 current sensor + magnetometer compensation system and benchmarking it against real drone flight data, providing a reproducible low-cost validation absent from current literature.

7. METHODOLOGY

7.1 Overall Methodology

The overall methodology adopted for the proposed magnetic field compensation system is presented in Figure 7.1. The complete workflow was designed in a systematic manner to study the magnetic field disturbance generated by the motor and Electronic Speed Controller (ESC), and to develop an effective compensation technique for improving magnetometer accuracy. The methodology includes several stages such as component selection, circuit design, hardware assembly, sensor calibration, data acquisition, compensation processing, and final output generation.

Figure 7.1 Flowchart of the Proposed Magnetic Field Compensation System



Initially, suitable components were selected based on performance, compatibility, and ease of integration. The major components used in the system include a microcontroller for processing, a magnetometer sensor for magnetic field measurement, a current sensor for motor current monitoring, a brushless DC motor, an ESC for motor speed control, and a regulated power supply. The selected components were chosen to ensure reliable real-time operation and accurate sensor interfacing.

After component selection, the circuit design stage was carried out to establish proper electrical connections between all modules. The magnetometer sensor was interfaced with the microcontroller through I²C communication, while the current sensor output was connected through an analog input channel. The ESC was controlled using PWM signals generated by the microcontroller. Proper grounding and supply connections were maintained to ensure stable system performance and reduce unwanted noise.

Once the circuit design was completed, the hardware assembly was performed by physically integrating all components according to the developed schematic. Mechanical mounting and wire routing were arranged carefully to avoid loose connections and minimize electromagnetic interference caused by power lines and motor wiring.

The next stage involved sensor calibration. Magnetometer calibration was performed to remove bias errors and improve measurement consistency. Similarly, the current sensor was calibrated to ensure accurate conversion of analog readings into current values. This calibration process was essential for obtaining reliable input data for the compensation model.

Following calibration, real-time data acquisition was carried out under different motor operating conditions. Magnetic field readings from the magnetometer and current values from the current sensor were continuously sampled by the microcontroller and transmitted for monitoring and analysis. The collected data clearly showed that motor current variations caused disturbances in the magnetic field measurements.

To eliminate these disturbances, the acquired data was processed using the proposed compensation algorithm. Initially, baseline offset removal was applied to remove constant bias from the magnetic field readings. Then, linear current-based compensation was implemented to estimate and subtract motor-induced interference. Where necessary, a non-linear term was also considered for higher-order effects. Finally, a smoothing filter was applied to reduce disturbance and transient fluctuations in the signal.

The final stage of the methodology produces a compensated and stabilized magnetic field output. This corrected output provides improved accuracy and consistency compared to raw sensor data, making it suitable for real-time embedded applications such as drone navigation, orientation sensing, and motor-driven control systems.

Thus, the proposed methodology provides a complete framework for implementing magnetic field compensation from hardware setup to final signal enhancement.

8. ELECTROMAGNETIC INTERFERENCE (EMI) MITIGATION AND COMPENSATION TECHNIQUES

8.1 EMI INTRODUCTION

Electromagnetic Interference (EMI) refers to unwanted electromagnetic signals that disturb the normal operation of electronic devices. EMI can affect sensors, communication systems, and control circuits in electronic systems such as drones. Understanding EMI and applying proper mitigation techniques helps improve system reliability and performance.

8.2 TYPES OF EMI

EMI can be classified based on how the interference is generated and transmitted.

Conducted EMI:

Conducted EMI travels through electrical conductors such as power cables or signal wires. It commonly occurs when noise from motors or power supplies travels through connected circuits.

Radiated EMI:

Radiated EMI spreads through the air in the form of electromagnetic waves. Devices such as motors, transmitters, and switching circuits can radiate electromagnetic energy that interferes with nearby electronics.

Capacitive Coupling:

Capacitive coupling occurs when electric fields between nearby conductors induce unwanted voltage in adjacent circuits.

Inductive Coupling:

Inductive coupling happens when changing magnetic fields induce unwanted currents in nearby conductive loops.

8.3 EMI IN DRONE TESTING

Electromagnetic Interference (EMI) is a critical issue in modern drone systems due to the presence of high-frequency electronic components such as motors, Electronic Speed Controllers (ESCs), communication modules, and power distribution units. These

components generate electromagnetic noise that can interfere with sensitive subsystems like GPS, sensors, and flight controllers.

EMI can lead to signal distortion, inaccurate sensor readings, communication failure, and overall instability of the drone. Therefore, it is essential to implement effective EMI mitigation techniques during both the design and testing phases of drone development.

8.3.1 SOURCES OF EMI IN DRONE SYSTEMS

In drone systems, EMI can originate from multiple sources, including:

- Brushless DC motors generating switching noise
- ESCs producing high-frequency signals
- Power distribution circuits
- Wireless communication modules (RF transmitters)
- Long connecting wires acting as antennas

8.4 EMI MITIGATION TECHNIQUES (HARDWARE LEVEL)

Electromagnetic Interference (EMI) in drone systems can be reduced using two main approaches: Passive techniques and Active techniques. Passive methods focus on physical design and hardware improvements, while active methods involve signal processing and dynamic control.

8.4.1 Passive EMI mitigation techniques

Passive EMI mitigation techniques are implemented through hardware design and do not require external power or control systems. These techniques are simple, reliable, and widely used in drone systems.

Passive techniques focus on layout, wiring, and grounding improvements:

- **Wiring separation:** High-power cables are separated from signal lines to reduce noise coupling
- **Twisted wires:** Reduces electromagnetic radiation by minimizing loop area
- **Short signal paths:** Minimizes noise pickup
- **Proper grounding:** Avoids ground loops using star grounding configuration



Figure 8.1 proper star pattern wiring and Ferrite bead

- **Physical separation:** High EMI components (motors, pumps) are placed away from sensors
- **Ferrite beads & filters:** Used to suppress high-frequency noise

These techniques significantly reduce interference without adding complex electronics.

8.4.2 Active EMI mitigation techniques

Active EMI mitigation techniques involve dynamic methods and signal processing approaches to reduce interference during operation.

When passive methods are insufficient, active methods are used:

- **Ground isolation:** Separates noisy power circuits from sensitive electronics
- **Optocouplers (opto-isolators):** Transfer signals without electrical connection
- **Signal isolation circuits:** Prevent noise propagation

These techniques directly block EMI transmission paths and improve signal quality in drone systems.

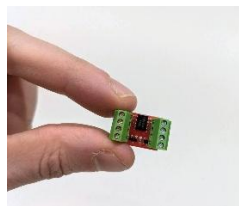


Figure 8.2 The sparkfun optocoupler

8.5 COMPENSATION TECHNIQUES (SOFTWARE LEVEL)

8.5.1 Magnetic Field Distortion In Drones

Even after EMI mitigation, magnetometer readings are still affected by onboard magnetic sources. These distortions are classified as:

- **Hard iron effect:** Constant offset caused by permanent magnetic fields
- **Soft iron effect:** Distortion due to surrounding materials altering field shape

These distortions cause the measured magnetic field to deviate from the ideal spherical distribution.

8.5.2 Hard And Soft Iron Compensation Method

The most commonly used compensation technique in drones is **hard and soft iron calibration**, as described by Analog Devices.

In this method:

- Hard iron error is removed by subtracting offset
- Soft iron error is corrected using scaling and rotation
- Distorted ellipsoid data is transformed into a sphere

This method is widely implemented in flight controllers but has limitations:

- Requires manual calibration
- Assumes constant interference
- Not effective for dynamic conditions

8.5.3 Ellipsoid Fitting-Based Compensation

An improved method uses **ellipsoid fitting**, developed in research from Stanford University.

Steps involved:

- Collect magnetometer data in multiple orientations
- Fit the data to an ellipsoid model
- Apply transformation to obtain a sphere

This method corrects:

- Offset errors
- Scale factor errors
- Axis misalignment

However:

- Requires large datasets
- Needs full drone rotation
- Computationally complex

8.5.4 Optimization-Based Compensation Methods

Advanced methods use optimization algorithms such as **Levenberg–Marquardt**:

- Builds a mathematical model of distortion
- Iteratively estimates calibration parameters
- Improves accuracy of heading estimation

Limitations:

- High computational cost
- Not suitable for real-time embedded drone systems

8.5.5 Sensor Fusion-Based Compensation

In drone systems, magnetometer data is often combined with:

- Gyroscope
- Accelerometer

Using filtering techniques such as Kalman filters:

- Reduces noise
- Improves stability
- Enhances heading estimation

However:

- Cannot fully remove magnetic interference
- Depends on multiple sensors

8.5.6 Ai And Data-Driven Compensation Methods

Recent approaches use machine learning:

- Learns interference patterns
- Handles nonlinear distortions
- Improves accuracy in complex environments

Limitations:

- Requires training data
- High processing requirements
- Difficult for real-time drone systems

9.COMPONENTS SELECTION

9.1 INTRODUCTION

The selection of appropriate components is a crucial step in the design and development of the proposed system. Each component is chosen based on its performance, compatibility, efficiency, cost-effectiveness, and suitability for the application. This chapter describes the components used in the system along with the reasons for their selection.

9.2 LIST OF COMPONENTS

The major components used in this project are:

- Arduino Nano (ATmega328P)
- Magnetometer (HMC5883L)
- Current Sensor (ACS712)
- Buck Converter
- Li-Po Battery (3S, 11.1V, 2200mAh)
- BLDC Motor (A2212)
- Electronic Speed Controller (30A SimonK ESC)

9.3 DESCRIPTION OF COMPONENTS

9.3.1 Arduino Nano (ATmega328P)

Description:

Arduino Nano is a compact microcontroller board based on the ATmega328P. It is widely used for embedded applications due to its small size, ease of programming, and sufficient number of input/output pins.

Reason for Selection:

The Arduino Nano was selected as the primary microcontroller for this project. Compared to the Arduino Uno, the Nano offers the same ATmega328P processing capability in a significantly smaller form factor, making it more suitable for compact sensor-integration setups where board real estate is limited. While the ESP32 provides Wi-Fi/Bluetooth connectivity and dual-core processing, it introduces unnecessary complexity, higher power consumption, and a steeper firmware development curve for a project that only requires basic

I2C/analog data acquisition. The Nano's native I2C support, 5V compatibility with all sensors used, and ease of programming via Arduino IDE made it the optimal choice.

9.3.2 Magnetometer (QMC5883L)

Description:

The QMC5883L is a 3-axis digital magnetometer used to measure the strength and direction of magnetic fields. It is commonly used for compass applications.

Reason for Selection:

The HMC5883L was chosen as the magnetometer for detecting and compensating magnetic field disturbances generated by the BLDC motor and ESC. Compared to the MPU-9250 (which integrates a gyroscope, accelerometer, and magnetometer in one package), the HMC5883L is a dedicated 3-axis magnetometer with higher magnetic sensitivity (up to ± 8 Gauss range) and a simpler I2C interface, reducing both processing overhead and cost. Since this project requires only magnetic field measurement without inertial sensing, using the MPU-9250's full IMU capability would be over engineering. The HMC5883L's lower cost, dedicated function, and straightforward integration with the Arduino Nano made it the preferred choice..

9.3.3 Current Sensor (ACS712)

Description:

ACS712 is a Hall-effect based current sensor used to measure AC and DC current. It provides an analog voltage output proportional to the current flowing through it.

Reason for Selection:

The ACS712 Hall-effect current sensor was selected over the INA219 shunt-based current sensor. The ACS712 provides galvanic isolation between the high-current motor circuit and the microcontroller, which is critical for safety when measuring currents from a 30A ESC and BLDC motor. In contrast, the INA219 requires a shunt resistor in-line with the load and lacks isolation, posing a risk of ground loop interference and potential damage to the Arduino in high-current environments. The ACS712 (30A variant) also directly outputs an analog voltage proportional to current, requiring no additional conversion hardware, and is readily compatible with the Nano's ADC pins.

9.3.4 Buck Converter

Description:

A buck converter is a DC-DC step-down converter used to reduce voltage from a higher level to a lower level efficiently.

Reason for Selection:

A DC-DC buck converter was used to step down the 11.1V LiPo battery voltage to the 5V required by the Arduino Nano and sensors, rather than using a linear voltage regulator such as the LM7805. Linear regulators dissipate excess voltage as heat — at 11.1V input and 5V output with a 500mA load, an LM7805 would dissipate approximately 3W as heat, requiring a heatsink and reducing overall system efficiency to around 45%. The buck converter, by contrast, achieves efficiencies of 85–95%, generates minimal heat, and extends battery life significantly — a critical consideration given the limited capacity of the 1200mAh LiPo pack.

9.3.5 Li-Po Battery (3S, 11.1V, 2200mAh)

Description:

A Lithium Polymer (Li-Po) battery is a rechargeable battery known for its high energy density and lightweight properties. A 3S battery provides a nominal voltage of 11.1V.

Reason for Selection:

A 3S Lithium Polymer (LiPo) battery was chosen over a comparable NiMH pack. LiPo batteries offer a significantly higher energy density (approximately 150–200 Wh/kg vs. 60–120 Wh/kg for NiMH), meaning a lighter and more compact pack can deliver the same energy. For driving a BLDC motor via a 30A ESC, LiPo also provides a high discharge rate (C-rating), ensuring sufficient instantaneous current delivery without voltage sag. NiMH packs of equivalent capacity would be heavier, bulkier, and would exhibit higher internal resistance leading to more pronounced voltage drops under load.

9.3.6 BLDC Motor (A2212)

Description:

The A2212 is a brushless DC motor commonly used in drones for propulsion. It offers high efficiency and high-speed operation.

Reason for Selection:

The A2212 Brushless DC (BLDC) motor was selected over a conventional brushed DC motor. BLDC motors eliminate physical brush-commutator contact, resulting in lower friction, longer operational life, and reduced electromagnetic interference (EMI). This is especially important in this project since minimizing unintended EMI sources aids the magnetic field compensation study. Brushed DC motors generate significant sparking at the commutator, which creates both electrical noise and additional magnetic field disturbances — counterproductive to the project's goals. The A2212 also provides a superior power-to-weight ratio, making it the standard choice in similar embedded motor applications..

9.3.7 Electronic Speed Controller (30A SimonK ESC)

Description:

An Electronic Speed Controller (ESC) is used to control the speed of a BLDC motor by regulating the power supplied to it. The 30A SimonK ESC is designed for fast response and smooth control.

Reason for Selection:

The 30A SimonK ESC was chosen over a generic PWM-based ESC. The SimonK firmware is specifically optimized for multi-rotor/BLDC motor control, offering faster PWM refresh rates and more precise throttle response. A generic ESC with default firmware typically uses lower refresh rates (~50Hz), introducing throttle lag and less consistent current draw patterns. Since this project studies the magnetic field generated during motor operation, consistent and controllable motor behavior is essential for repeatable measurements. The SimonK ESC's current handling (30A continuous) also comfortably exceeds the A2212's operating current, providing a safe margin.

10. CIRCUIT DESIGN

10.1 Circuit Overview

The circuit design and validation of a Brushless DC (BLDC) motor drive system designed to demonstrate controlled motor operation using an Arduino Nano microcontroller, a 30A Electronic Speed Controller (ESC), and supporting sensor modules. The entire circuit was designed and validated using the Wokwi online electronics simulator, which provides a realistic virtual prototyping environment before physical hardware assembly.

Wokwi is a free, browser-based simulator that supports Arduino, ESP32, and a wide range of electronic components. It allowed us to verify the logic, pin connections, and sensor readings in a safe virtual environment, reducing the risk of hardware damage during development.

The BLDC motor drive system consists of a power stage and a control stage. The power stage handles high-current motor drive through the ESC, while the control stage manages PWM signals, current sensing, and compass heading data via the Arduino Nano.

10.2 Block Diagram Summary

The system follows this signal and power flow:

1. LiPo Battery (3S 11.1V) supplies main power to ESC and buck converter.
2. Toggle switch acts as a safety cutoff on the negative power line.
3. ESC receives PWM signal from Arduino D10 and drives the BLDC motor.
4. ACS712 current sensor monitors motor current and reports to Arduino A0.
5. HMC5883L magnetometer communicates with Arduino via I2C (SDA/SCL).
6. Buck converter steps 11.1V down to 5V to power Arduino and sensors.

Figure 10.1 Circuit Diagram Of BLDC Motor Drive

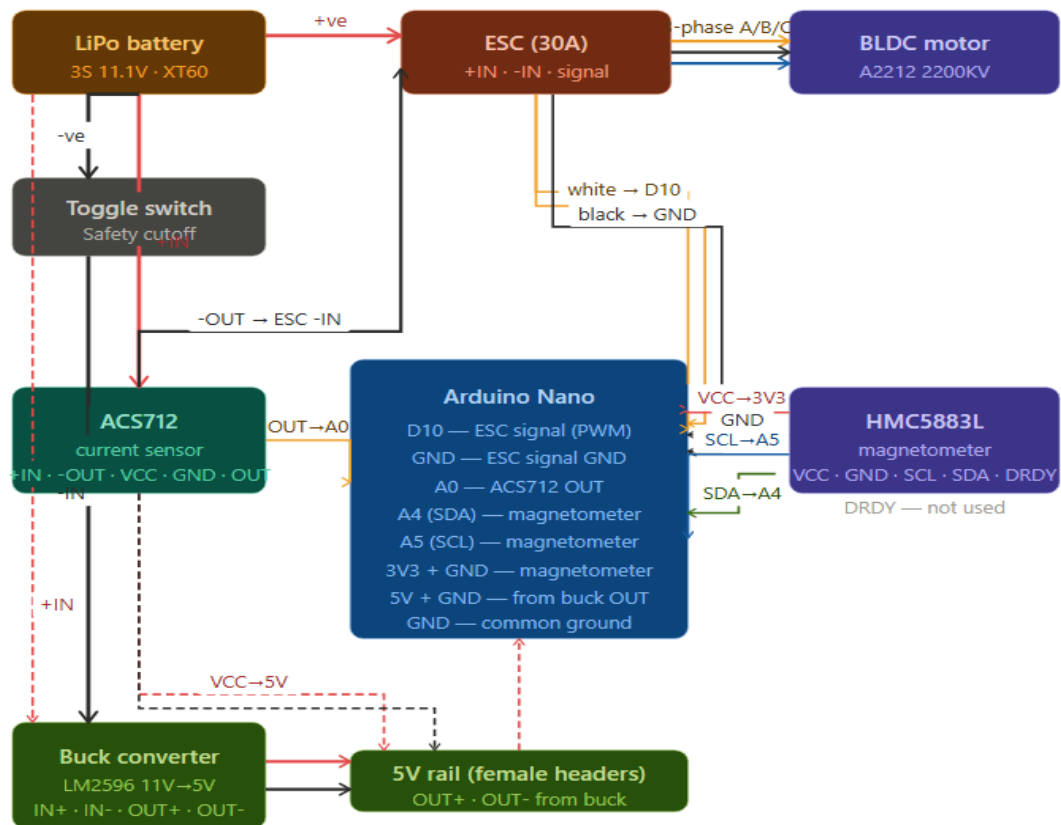


Table 10.1 Wire Colour Legend

Wire Colour	Description
Red (thick)	High-current +ve: Battery → ACS712 → ESC +IN
Black (thick)	High-current -ve GND: Battery → Toggle Switch → ESC -IN
Yellow	3-phase motor wires: ESC → BLDC Motor (A/B/C phases) / I2C SCL
Red (dashed thin)	5V regulated power: Buck OUT → Arduino → Sensors
Green	Signal wires: ACS712 OUT → A0 / Buck → 5V Rail
Cyan / Blue	3.3V and I2C SDA (A4) → HMC5883L Magnetometer
White	ESC signal wire: ESC → Arduino D10 (PWM)

10.3 Circuit validation

The designed circuit connections were initially obtained using AI-assisted tools to ensure proper interfacing between all components. To verify the correctness of these connections, the design was cross-checked with the reference configurations available in Wokwi.

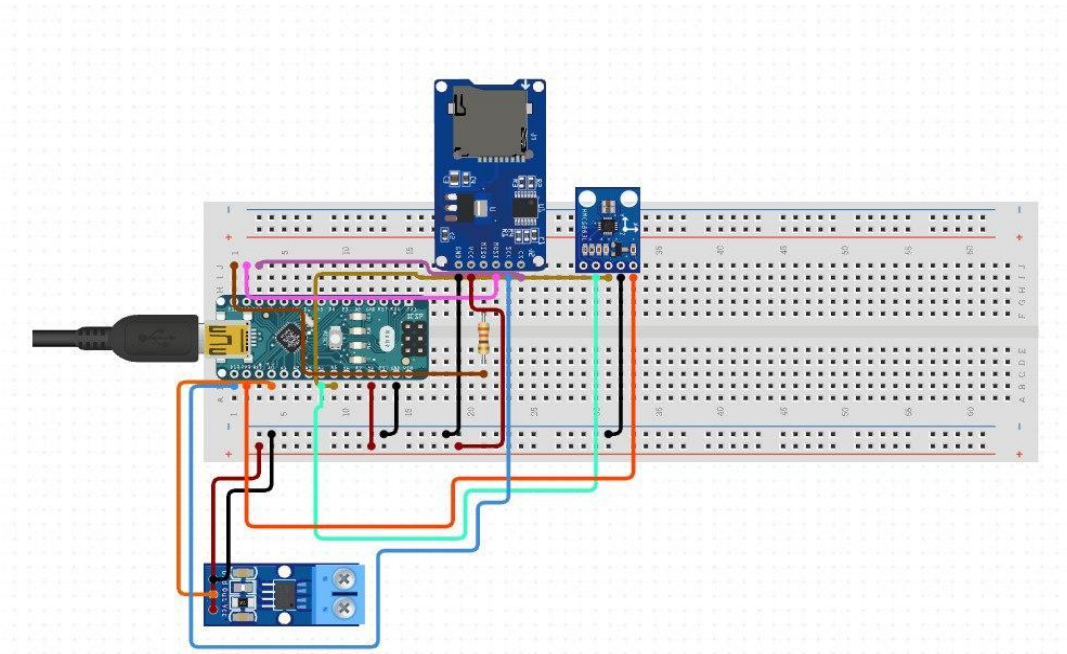


Figure 10.2 online circuit design from wokwi

In this process, the connections of key components such as the microcontroller, magnetometer sensor, and current sensor were carefully compared with standard configurations provided in the platform. The I²C communication lines and analog input connections were found to be consistent with the expected interfacing methods, confirming the correctness of the sensor integration.

However, complete validation of the motor drive section was not performed within the platform. When attempting to include the BLDC motor and Electronic Speed Controller (ESC), compatibility limitations were observed, particularly involving differences in supported microcontroller configurations. As a result, the motor-ESC interface could not be fully verified using the tool.

Therefore, the validation using Wokwi was limited to approximately 50% of the circuit, mainly focusing on sensor interfacing and control connections. The remaining motor drive section was validated through practical hardware implementation and experimental testing.

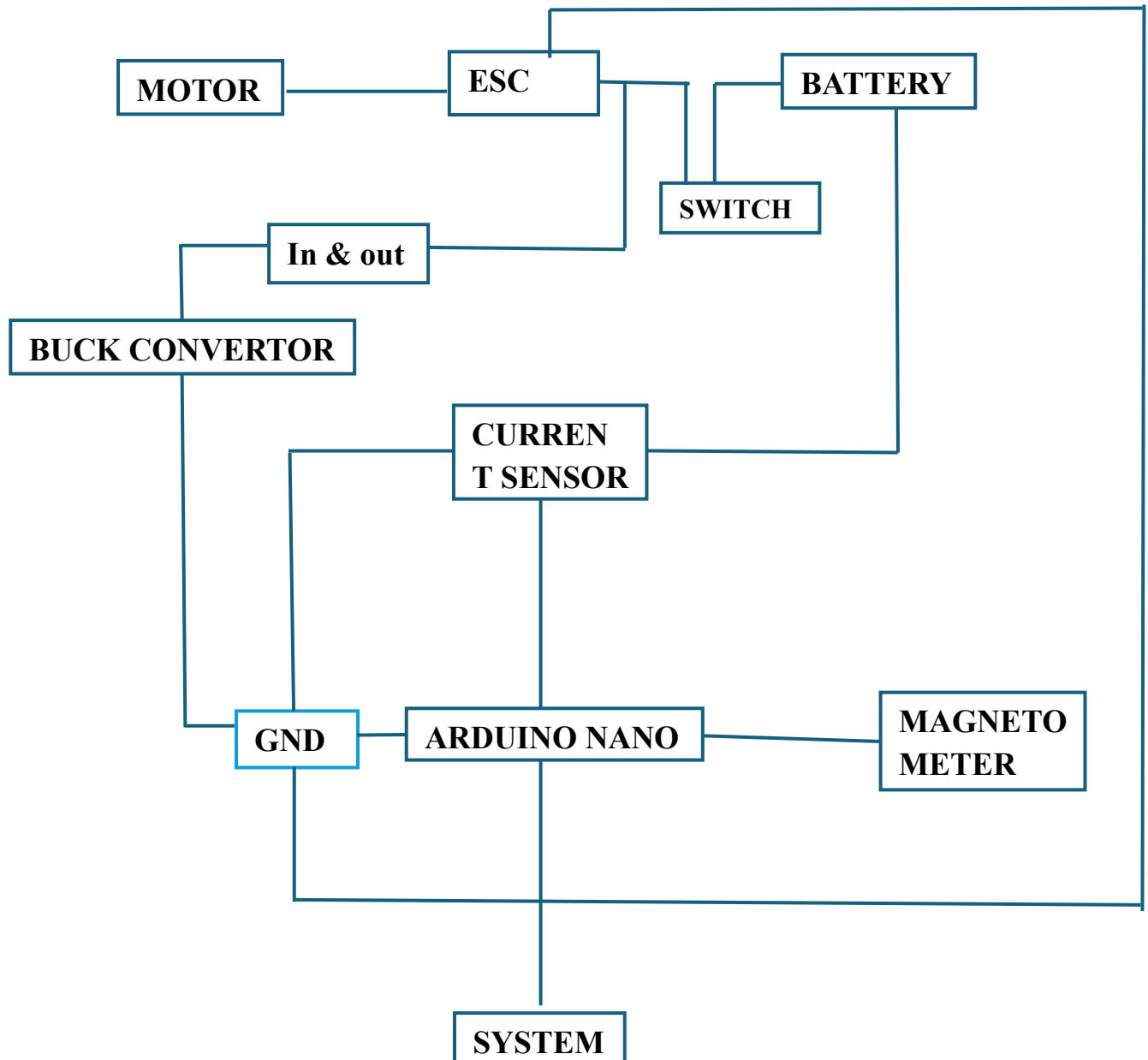
Despite this limitation, the cross-verification process helped ensure that the critical sensing and data acquisition connections were accurate, thereby reducing the possibility of wiring errors during hardware assembly.

11.SYSTEM ARCHITECTURE

11.1 System Architecture Overview

The system consists of sensing, processing, and actuation units. The magnetometer detects magnetic field variations, the current sensor measures motor current, and the embedded controller processes the data to compensate interference.

11.2 Block Diagram



11.3 Dataflow Architecture

11.3.1 Sensor Data Acquisition

Sensor data acquisition represents the primary stage of the proposed system, in which raw measurements are obtained from the magnetometer and current sensing module. The magnetometer measures the magnetic flux density components along the three orthogonal axes, namely X , Y , and Z , while the current sensor monitors the instantaneous current consumed by the motor. These sensor outputs are continuously sampled by the microcontroller using appropriate interfacing protocols.

Time(ms)	Current(A)	RawX	RawY	RawZ
1	6.99	-0.09	-0.09	-0.1
234	7.09	-0.09	-0.09	-0.1
438	7.12	-0.09	-0.09	-0.1
641	7.17	-0.09	-0.09	-0.1
844	7.23	-0.09	-0.09	-0.1
1047	7.23	-0.09	-0.09	-0.1
1250	7.28	-0.09	-0.09	-0.1
1454	7.3	-0.09	-0.09	-0.1
1656	7.33	-0.09	-0.09	-0.1
1860	7.36	-0.09	-0.09	-0.1
2063	7.38	-0.09	-0.09	-0.1
2267	7.41	-0.09	-0.09	-0.1
2469	7.44	-0.09	-0.09	-0.1
2673	7.44	-0.09	-0.09	-0.1
2876	7.46	-0.09	-0.09	-0.1
3079	7.49	-0.09	-0.09	-0.1
3282	7.52	-0.09	-0.09	-0.1
3485	7.54	-0.09	-0.09	-0.1
3689	7.54	-0.09	-0.09	-0.1
3892	7.62	-0.09	-0.09	-0.1
4096	7.33	-0.09	-0.09	-0.1
4298	7.38	-0.09	-0.09	-0.1
4502	7.44	-0.09	-0.09	-0.1

Figure 11.1 Raw magnetometer measurements

11.3.2 Baseline Offset Removal

Baseline offset removal is implemented to eliminate the constant bias component present in the sensor measurements. Even under no-load or idle operating conditions, the magnetometer may exhibit non-zero outputs due to ambient magnetic field interference, sensor drift, and inherent hardware imperfections. To mitigate this effect, multiple sensor samples are acquired during the motor's inactive state, and the mean value of each axis is computed to estimate the corresponding offset component.

The estimated offset values for the X , Y , and Z axes are subsequently subtracted from the raw sensor measurements according to the following relation:

$$r_x = x - \text{offset_X}$$

$$r_y = y - \text{offset_Y}$$

$$r_z = z - \text{offset_Z}$$

Where,

- r_x, r_y, r_z - offset corrected values
- x, y, z - represents raw magnetometer data
- $\text{offset_x}, \text{offset_y}, \text{offset_z}$ - offset values.

This removal process ensures that the processed sensor data accurately represents only the dynamic variations in the magnetic field induced by motor operation and electromagnetic interference.

11.3.3 Linear Compensation

Linear compensation is employed to mitigate the magnetic interference component that exhibits a direct proportionality to the motor current. During motor and Electronic Speed Controller (ESC) operation, the flow of current generates electromagnetic fields that introduce disturbances in the magnetometer measurements. This interference component is modeled under the assumption of a linear relationship between the measured current and the induced magnetic field disturbance.

Accordingly, a proportional compensation term is subtracted from the baseline offset-corrected sensor outputs for each axis using the following relations:

$$C_x = r_x - (K_x * I)$$

$$C_y = r_y - (K_y * I)$$

$$C_z = r_z - (K_z * I)$$

Where,

- C_x, C_y, C_z - final compensated values
- r_x, r_y, r_z - offset corrected values (magnetometer data)
- K_x, K_y, K_z - linear coefficient
- I - current.

This compensation technique effectively suppresses the current-induced magnetic interference component, thereby improving the accuracy and reliability of the magnetometer measurements.

11.3.4 Non-Linear Compensation

Non-linear compensation is incorporated to compensate higher-order magnetic interference components that cannot be accurately modeled using a linear current-dependent relationship. In practical embedded motor-drive systems, phenomena such as ESC switching transients, magnetic core saturation, hysteresis effects, and electromagnetic coupling can introduce non-linear distortions in the magnetometer measurements. To compensate for these effects, a second-order current-dependent term is introduced into the compensation model, expressed as:

$$C_x = r_x - (K_x * I) - (K_2 * I^2)$$

$$C_y = r_y - (K_y * I) - (K_2 * I^2)$$

$$C_z = r_z - (K_z * I) - (K_2 * I^2)$$

Where,

- C_x, C_y, C_z - final compensated values
- r_x, r_y, r_z - offset corrected values (magnetometer data)
- K_x, K_y, K_z - linear coefficient
- I - current.
- K_2 - Nonlinear coefficient

11.3.5 Smoothing Filter

A smoothing filter is implemented to transient fluctuations present in the compensated magnetic field measurements. In practical sensor-based systems, magnetometer outputs are often affected by magnetic field interference, measurement uncertainty, and external environmental disturbances, resulting in minor variations in the acquired data.

To reduce these unwanted fluctuations, a first-order Exponential Moving Average (EMA) filter is applied to the compensated signal. The filtering operation is represented as:

$$C_x = (0.8 * \text{previous value}) + (0.2 * \text{current value})$$

$$C_y = (0.8 * \text{previous value}) + (0.2 * \text{current value})$$

$$C_z = (0.8 * \text{previous value}) + (0.2 * \text{current value})$$

Where,

- 0.8 is used for stability
- 0.2 for responsiveness

11.3.6 Outcome

The implemented system effectively measures and compensates motor and ESC-induced magnetic interference. Offset correction and linear compensation stabilize the magnetic field readings, making them independent of current variations. A smoothing filter further improves data stability by minimizing magnetic field. The results confirm that linear compensation is sufficient, providing accurate and reliable measurements for practical applications

11.4 Working Process

- The **battery** provides the main DC power supply to the complete system.
- A **switching unit** controls the ON/OFF operation of the motor drive circuit.
- The **ESC (Electronic Speed Controller)** receives power from the battery and regulates motor speed using PWM control signals.
- The **motor** operates based on the control output from the ESC and generates magnetic interference during rotation.
- A **current sensor** is connected in series with the motor supply line to measure real-time motor current.
- The **buck converter** steps down battery voltage to a regulated DC level for powering the Arduino and sensors.
- The **magnetometer sensor** measures magnetic field values along the X, Y, and Z axes.
- The **Arduino Nano** collects data from both the current sensor and magnetometer for processing.
- A **linear compensation algorithm** is applied to remove magnetic interference proportional to motor current.

- **Offset correction and smoothing filter techniques** are used to improve measurement accuracy and reduce magnetic field.
- The final output provides **stable and compensated magnetic field readings** suitable for practical applications such as drone navigation and magnetic sensing.

12. EXPERIMENTAL SECTION

12.1 Experimental setup Description

The experimental setup was designed to quantify, model, and compensate the magnetic interference generated by a brushless direct current (BLDC) motor and its associated electronic speed controller (ESC) on a co-located three-axis magnetometer, using a low-cost embedded system based on the Arduino Nano microcontroller. The entire system was assembled on a perfboard to provide a controlled, magnetically benign test environment during ground-based testing.

The primary interference source was an A2212/1000KV three-phase BLDC motor, driven by a SimonK-firmware 30A ESC. The ESC received pulse-width modulation (PWM) throttle commands from digital pin D10 of the Arduino Nano, with pulse widths ranging from 1000 μ s (minimum throttle/armed idle) to 2000 μ s (maximum throttle) at a fixed frequency of 50 Hz. The motor was operated under no-load conditions throughout all experiments to ensure that the measured magnetic deviation was attributable solely to the electromagnetic emissions of the spinning motor, the switching behaviour of the ESC, and the pulsed stator currents, rather than any mechanical loading effect.

The QMC5883L triaxial magnetometer was interfaced with the Arduino Nano via the I²C bus (SDA on pin A4, SCL on pin A5) at a device address of 0x1E. The sensor was configured with a gain setting of 0 (measurement range ± 1.3 Gauss) and a data output rate of 100 Hz. The magnetometer was mounted in a fixed position on the perfboard, deliberately placed at a proximity to the motor system to ensure that the interference field was detectable and measurable, thereby mimicking the constrained physical layout found in compact UAV platforms where physical separation between sensor and motor is inherently limited.

Motor current was monitored in real time using an ACS712-30A Hall-effect current sensor inserted in series with the main positive power line between the LiPo battery and the ESC. The sensor outputs an analog voltage proportional to the current flowing through it (sensitivity: 66 mV/A, midpoint voltage: 2.5V at zero current), which was digitised by the 10-bit ADC on analog pin A0 of the Arduino Nano at a sampling rate of 100 Hz, synchronised with the magnetometer acquisition. This simultaneous sampling of current and magnetic field at the same rate was fundamental to the derivation of the linear compensation model.

Power was supplied by a 3-cell (3S) lithium polymer (LiPo) battery rated at 11.1V nominal and 2200mAh capacity with an 80C discharge rating, connected via an XT60 connector. An LM2596-based buck converter stepped the 11.1V battery voltage down to a regulated 5V supply rail, which powered the Arduino Nano, the ACS712 current sensor, and provided the 3.3V regulated line (via the Arduino's onboard regulator) to the QMC5883L magnetometer. A toggle switch was inserted in the negative power rail between the battery and the circuit to serve as a hardware safety cutoff, preventing unintended motor arming.

All firmware was developed in C++ using the Arduino IDE (v2.x) and uploaded to the ATmega328P microcontroller over USB. The firmware incorporated the ESC arming sequence, PWM throttle control, I²C magnetometer acquisition, ADC current measurement, hard-iron offset correction, soft-iron scaling, and the linear current-based compensation model. Data from both sensors were streamed in real time via the serial monitor at 9600 baud and recorded to a comma-separated values (CSV) file through a serial-to-CSV logging utility for subsequent analysis in Microsoft Excel.

12.2 Hardware Connection

The hardware connections were designed to establish two distinct electrical paths: a high-current power path carrying the motor drive current from the battery through the current sensor and ESC to the BLDC motor, and a low-voltage signal path carrying data signals between the sensors and the Arduino Nano microcontroller.

12.2.1 High-current Power Path

The positive terminal of the 3S LiPo battery was connected to the +IN terminal of the ACS712 current sensor via a thick-gauge red wire, ensuring that the entirety of the motor drive current passed through the Hall-effect sensing element. The –OUT terminal of the ACS712 was then connected to the positive input terminal of the ESC, forming the forward current path. The negative terminal of the battery was routed through the SPST toggle safety switch before connecting to both the negative input of the ESC and the IN– terminal of the LM2596 buck converter. The three-phase output wires of the ESC (carrying the commutated motor drive waveforms) were connected directly to the three phase terminals of the A2212 BLDC motor.

A secondary tap from the battery positive terminal was connected to the IN+ terminal of the buck converter. The buck converter was adjusted to output a stable 5V supply, which was distributed via female header pins to power the ACS712, the Arduino Nano (via its VIN or

5V pin), and to feed the Arduino's onboard 3.3V linear regulator for the magnetometer supply.

12.2.2 Signal Path Connections

The ESC carries two thin-gauge signal wires in addition to its power wires. The white signal wire was connected to digital pin D10 of the Arduino Nano, which was configured as a PWM output generating 50 Hz throttle pulses with widths between 1000 μ s and 2000 μ s. The black thin wire was connected to a GND pin on the Arduino Nano to establish a common signal ground reference between the ESC and the microcontroller.

The QMC5883L magnetometer was connected to the Arduino Nano over the I²C bus. The SDA line was connected to analog pin A4 (which serves as the I²C data line on the ATmega328P) and the SCL line was connected to analog pin A5 (the I²C clock line). The VCC pin of the magnetometer was supplied from the Arduino Nano's 3.3V regulated output pin, and its GND pin was connected to the Arduino's ground. The DRDY (data ready) pin was left unconnected as polling mode was used.

The ACS712 current sensor's signal output pin (OUT) was connected to analog pin A0 of the Arduino Nano. The VCC pin of the ACS712 was connected to the regulated 5V rail and its GND pin to the common ground rail. The ADC on pin A0 sampled the analog voltage output of the sensor at 100 Hz, with the reading converted to a current value in amperes using the formula:

$$I (A) = (ADC_reading - 512) \times (5.0 / 1024) / 0.066$$

where 512 represents the zero-current midpoint ($V_{out} = 2.5V$), and 0.066 V/A is the ACS712 sensitivity constant.

Table 12.1 The Complete Pin-By-Pin Connection Map For The Experimental System.

From component / pin	To component / pin	Purpose
LiPo battery +ve (XT60)	ACS712 +IN terminal	High-current positive feed through sensor
ACS712 –OUT terminal	ESC +IN	Positive power to ESC after current measurement
LiPo battery –ve (XT60)	Toggle switch IN	Negative power rail through safety cutoff

Toggle switch OUT	ESC –IN and buck converter IN–	Common negative rail to ESC and buck converter
LiPo battery +ve (tap)	Buck converter IN+	11.1V input to buck converter
Buck converter OUT+	Female header +5V rail	Regulated 5V distribution to sensors and Arduino
Buck converter OUT–	Female header GND rail	Regulated GND rail
ESC 3-phase wires (A/B/C)	BLDC motor 3 terminals	3-phase commutation drive to motor
ESC thin white wire (signal)	Arduino Nano D10	PWM throttle signal output (1000–2000 μ s, 50 Hz)
ESC thin black wire (GND)	Arduino Nano GND	Signal ground reference
ACS712 VCC	5V rail (buck OUT+)	5V power supply to current sensor
ACS712 GND	GND rail (buck OUT–)	Ground reference for current sensor
ACS712 OUT	Arduino Nano A0	Analog current signal (0–5V proportional to 0–30A)
HMC5883L VCC	Arduino Nano 3V3	3.3V regulated supply to magnetometer
HMC5883L GND	Arduino Nano GND	Ground reference for magnetometer
HMC5883L SCL	Arduino Nano A5	I ² C clock line (100–400 kHz)
HMC5883L SDA	Arduino Nano A4	I ² C data line (bidirectional)

12.3 Sensor Calibration

Prior to conducting the motor interference experiments, each sensor and the motor-ESC system was individually calibrated to ensure accurate, bias-free measurements. All calibration routines were implemented as dedicated C++ functions within the Arduino IDE firmware and executed sequentially at system startup.

12.3.1 A2212 BLDC Motor and ESC Calibration

Before the motor could be controlled reliably, the ESC required a throttle range calibration to map its internal PWM acceptance window to the full 1000–2000 μs range generated by the Arduino Nano. The arming and calibration procedure was as follows:

- The Arduino firmware was programmed to output the maximum PWM pulse width of 2000 μs on pin D10.
- The 3S LiPo battery was then connected to the circuit with the toggle switch open; the switch was subsequently closed to power the ESC while the maximum throttle signal was active.
- The ESC emitted a series of audible confirmation tones indicating that it had registered the upper throttle endpoint.
- The firmware then immediately stepped the PWM signal down to 1000 μs (minimum throttle). The ESC emitted a further confirmation tone, completing the calibration and entering the armed-idle state.

Following the arming sequence, a no-load current profile was recorded by incrementally increasing the PWM signal from 1000 μs to 2000 μs in steps of 100 μs , holding each step for 30 seconds. The motor current at each throttle step was recorded via the ACS712 sensor, confirming a measurable current draw in the range of 0 to approximately 12 A across the full throttle range. The RPM response was also visually verified to increase monotonically with increasing PWM pulse width, confirming correct ESC calibration.

12.3.2 QMC5883L Magnetometer Calibration

Magnetometer calibration was performed with the motor completely powered off in order to establish a clean ambient magnetic baseline free from any motor-generated interference. The calibration procedure involved the following steps executed through the Arduino IDE firmware:

- I²C initialisation: The Wire library was used to establish communication with the HMC5883L at address 0x1E. Configuration registers were written to set a gain of 0 (± 1.3 Gauss range, resolution 0.92 mG/LSB) and a continuous measurement mode at 100 Hz.
- Idle-state sampling: With the motor powered off, 500 raw samples per axis (X, Y, and Z) were acquired at 100 Hz and stored in SRAM buffers.

- **Hard-iron offset computation:** The mean value of each axis buffer was computed to yield the hard-iron DC bias offsets: $\text{offset_X} = 1.593 \mu\text{T}$, $\text{offset_Y} = -0.042 \mu\text{T}$, $\text{offset_Z} = -2.678 \mu\text{T}$.
- **Offset correction:** All subsequent raw magnetometer readings were corrected by subtracting the corresponding offset: $r_x = x - \text{offset_X}$, $r_y = y - \text{offset_Y}$, $r_z = z - \text{offset_Z}$.
- **Soft-iron correction:** A 3×3 scaling matrix was estimated to correct for ellipsoidal distortion caused by soft-iron materials in the vicinity of the sensor, transforming the corrected ellipsoid response back into a spherical distribution.
- **Zero-mean verification:** The corrected outputs were verified to have a near-zero mean across all three axes, confirming successful bias removal.

Figure 12.1 QMC5883L calibration

12.3.3 ACS712 Current Sensor Calibration

- **ADC initialisation:** Analog pin A0 was configured with the internal 5V reference. The 10-bit ADC resolves 0–5V into 1024 levels, yielding a resolution of approximately 4.88 mV per LSB.
- **Zero-current baseline:** With the motor disconnected and the circuit powered, the ADC reading on A0 was recorded under zero-current conditions. The output voltage of the

ACS712 at zero current is nominally 2.5V (mid-rail), corresponding to an ADC reading of approximately 512. This value was stored as the zero-current reference in firmware.

- Linearity verification: The motor was run at increasing throttle levels and the ADC-derived current was compared against the expected linear response based on the 66 mV/A sensitivity specification of the ACS712. The measured current was found to increase linearly from 0 A at idle to approximately 12 A at maximum throttle under no-load conditions.
- Timestamp synchronisation: The ACS712 sampling loop was synchronised with the HMC5883L sampling loop to ensure that each current reading $I(t)$ was paired with a corresponding set of magnetometer readings (rx, ry, rz) at the same timestamp, enabling accurate correlation between current and magnetic deviation.

12.4 Experimental Procedure

12.4.1 Experimental Procedure

All components were assembled on a common perfboard. The Arduino Nano was mounted centrally, with the QMC5883L magnetometer, ACS712 current sensor module, and buck converter arranged around it. The motor and ESC were positioned adjacent to the board such that the motor's electromagnetic emissions would directly influence the magnetometer readings, replicating the interference conditions present in a compact UAV airframe. All wiring followed the connection map described in Section 10.2 and was secured with insulation tape to prevent short circuits.

The C++ firmware was developed in Arduino IDE (v2.x), compiled for the ATmega328P, and uploaded via USB. The firmware initialised all peripherals at startup in the following order:

- (i) Serial communication at 9600 baud for data logging
- (ii) I²C bus and HMC5883L sensor configuration
- (iii) ADC configuration for pin A0
- (iv) PWM timer configuration for pin D10 (50 Hz, 8-bit resolution)
- (v) ESC arming sequence execution.

12.4.2 Phase 1 — Baseline Data Collection and Offset Estimation

In the first phase, the developed code was uploaded to the microcontroller using the Arduino IDE platform. The system was operated without running the motor to capture the baseline magnetometer readings under idle conditions.

The sensor data, including magnetic field values along the three axes, was collected for a short duration and recorded in CSV (Comma-Separated Values) format. This dataset represents the inherent bias present in the magnetometer readings due to environmental factors and sensor characteristics.

The recorded data was then processed using Microsoft Excel to calculate the average values of each axis. These average values were considered as the baseline offset components (offset_X, offset_Y, offset_Z), which were later used to correct the raw sensor readings. The verified offset values were:

$$\text{offset_X} = 1.593 \mu\text{T}$$

$$\text{offset_Y} = -0.042 \mu\text{T}$$

$$\text{offset_Z} = -2.678 \mu\text{T}$$

These offsets represent the static hard-iron bias of the sensor within the assembled system, comprising contributions from the PCB traces, battery remnant field, and any permanently magnetised components in the vicinity of the sensor when the motor is not operating.

12.4.3 Phase 2 — Data Collection for Compensation Coefficient Estimation

In the second phase, the code was modified to include the calculated offset values, and the updated program was re-uploaded to the microcontroller. The system was then operated with the motor running at different throttle levels to observe the effect of current variation on the magnetic field.

During this phase, both magnetometer data and motor current values were collected continuously for a specified duration at each throttle condition. The data was recorded in CSV format for further analysis.

The collected dataset was analyzed using Microsoft Excel to establish the relationship between motor current and magnetic field variation. Based on this analysis, the linear

compensation coefficients (K_X , K_Y , K_Z) were determined for each axis. These coefficients represent the sensitivity of magnetic interference with respect to current.

Table 12.2 Throttle sweep profile

Steps	PWM(μ s)	Throttle(%)	Duration(s)
Step 1	1000	0	30
Step 2	1100	10	30
Step 3	1200	20	30
Step 4	1300	30	30
Step 5	1400	40	30
Step 6	1500	50	30
Step 7	1600	60	30
Step 8	1700	70	30
Step 9	1800	80	30
Step 10	1900	90	30
Step 11	2000	100	30

The total data collection phase spanned $11 \times 30 = 330$ seconds (5.5 minutes) for each complete throttle sweep. Multiple runs were conducted to verify the repeatability of the interference signatures across throttle levels.

12.4.4 Phase 3 - Implementation and Validation of Compensation Model

In the final phase, the estimated offset values and compensation coefficients were incorporated into the program, and the updated code was uploaded to the microcontroller.

The system was then operated in real time, and the compensated magnetic field output was visualized using the Serial Plotter available in the Arduino IDE. This allowed direct comparison between raw and compensated signals.

The results observed in the serial plotter demonstrated a noticeable reduction in magnetic field fluctuations after applying the compensation model. The compensated signal appeared smoother and more stable, confirming the effectiveness of the proposed approach.

12.5 Data Recording and Analysis

12.5.1 Data Acquisition via Serial Monitor

All sensor data were transmitted through the Arduino Nano's hardware UART at 9600 baud to a host computer running the Arduino IDE serial monitor. Each data frame transmitted to the serial monitor contained a comma-separated record of the following fields: timestamp

(milliseconds since boot), PWM command value (μs), motor current I in amperes, and the three offset-corrected magnetometer readings r_x , r_y , r_z in microtesla. The serial output was captured and saved as a comma-separated values (CSV) file for offline post-processing.

The CSV data were imported into Microsoft Excel for analysis. For each throttle step, the steady-state mean values of I , r_x , r_y , and r_z were extracted by averaging the readings recorded during the stable central portion of each 30-second window (discarding the first and last 5 seconds to eliminate transient settling effects). This yielded a dataset of 11 data points correlating mean motor current with mean magnetic deviation on each axis.

12.6 Mathematical Model

12.6.1 Magnetic Field Representation

The magnetic field measured by the onboard magnetometer can be expressed as the superposition of three distinct components:

$$B_{\text{measured}} = B_{\text{true}} + B_{\text{offset}} + B_{\text{interference}}$$

where B_{measured} is the magnetometer output, B_{true} is the actual ambient magnetic field, B_{offset} is a constant bias arising from sensor imperfections and static environmental factors, and $B_{\text{interference}}$ represents dynamic magnetic disturbances induced by the motors and electronic speed controllers (ESCs) during operation. The objective of the subsequent calibration procedure is to accurately estimate and subtract both B_{offset} and $B_{\text{interference}}$ from the measured signal, such that the corrected output closely approximates B_{true} . The mathematical framework used to model and isolate these components is developed in the following section.

12.6.2 Initial Mathematical Model

Proposed Formula

In the initial stage, the magnetic interference was assumed to be directly proportional to the motor current. Based on this assumption, the following compensation formula was proposed:

$$C = B_{\text{measured}} - (K \times I)$$

where C is the compensated magnetic field, K is the compensation coefficient, and I is the measured motor current.

Derivation

The derivation follows from the observation that magnetic disturbances generated by motors and ESCs scale with the current drawn during operation. Formally:

$$B_{\text{interference}} \propto I \Rightarrow B_{\text{interference}} = K \times I$$

Substituting into the field representation introduced in the previous section:

$$C = B_{\text{measured}} - B_{\text{interference}} = B_{\text{measured}} - (K \times I)$$

The compensation coefficient K was determined empirically from the recorded data, and the corrected output C was expected to approximate B_{true} .

Limitations and Failure Analysis

Despite the intuitive motivation behind this model, it failed to produce accurate results in practice. The primary reasons are as follows:

- The constant bias term B_{offset} was never removed. Since the model operated directly on the raw magnetometer output, the inherent sensor bias and static environmental offsets remained present in C .
- Even when the motor was completely off (i.e., $I=0$), the sensor readings exhibited a non-zero shift, confirming the presence of B_{offset} independent of motor activity.
- Because compensation was applied to already-biased data, the subtraction of $K \times I$ could not fully recover B_{true} , leading to residual errors that varied with operating conditions.

The initial model failed because it did not account for the static offset present in the sensor readings. Since this offset was not removed before applying the compensation, the output remained inaccurate. This highlighted the need for an improved model that first removes the baseline offset before performing interference compensation.

12.6.3 Improved Mathematical Model

The improved model addresses the limitations of the initial approach by introducing a three-stage processing pipeline: baseline offset removal, current-based interference compensation, and signal smoothing.

Stage 1 — Baseline Offset Removal

To eliminate the constant bias, the magnetometer is first sampled while the motor is in an idle state. The mean of these idle readings is computed along each axis to obtain the values offset_X , offset_Y , offset_Z . The offset-corrected output is then given by:

$$r_x = x - \text{offset_X}$$

$$r_y = y - \text{offset_Y}$$

$$r_z = z - \text{offset_Z}$$

where x , y , z are the raw magnetometer readings and r_x , r_y , r_z are the corrected values. This step ensures that the static bias is fully removed before any further processing.

Stage 2 — Interference Compensation

After offset removal, the residual interference is primarily caused by the magnetic field generated by motor current. Two compensation models were considered.

Linear Model:

The simplest assumption is that the interference scales linearly with current. The compensated output along each axis is:

$$C_x = r_x - (K_x * I)$$

$$C_y = r_y - (K_y * I)$$

$$C_z = r_z - (K_z * I)$$

where K_x , K_y , K_z are the axis-wise linear compensation coefficients and I is the measured motor current.

Non-Linear Model:

In practice, the relationship between current and magnetic interference may exhibit higher-order behaviour. To account for this, a quadratic term is introduced:

$$C_x = r_x - (K_x * I) - (K_2 * I^2)$$

$$C_y = r_y - (K_y * I) - (K_2 * I^2)$$

$$C_z = r_z - (K_z * I) - (K_2 * I^2)$$

Where, C_x , C_y , C_z are final compensated values and r_x , r_y , r_z are the offset corrected values (magnetometer data) and axis wise coefficient K_x , K_y , K_z , I is the measured motor current and K_2 captures the non-linear contribution. However, experimental results in this work demonstrated that the linear model provided sufficient accuracy, and the non-linear model was not required for the final implementation.

Stage 3 — Smoothing Filter

Even after compensation, the signal may still contain high-frequency noise and sudden fluctuations. A first-order Exponential Moving Average (EMA) filter is applied to improve signal stability:

$$C_x = (0.8 * \text{previous value}) + (0.2 * \text{current value})$$

$$C_y = (0.8 * \text{previous value}) + (0.2 * \text{current value})$$

$$C_z = (0.8 * \text{previous value}) + (0.2 * \text{current value})$$

In this implementation, a smoothing factor of $\alpha=0.8$ was selected, meaning the filter retains 80% of the previous output and incorporates 20% of the new measurement. This choice effectively suppresses high-frequency noise while maintaining a sufficiently fast response to real changes in the magnetic field, producing a stable and reliable output signal.

Final Combined Model

The complete processing pipeline can be expressed compactly as:

$$C_x = r_x - (K_x * I)$$

$$C_y = r_y - (K_y * I)$$

$$C_z = r_z - (K_z * I)$$

For the extended non-linear case:

$$C_x = r_x - (K_x * I) - (K_2 * I^2)$$

$$C_y = r_y - (K_y * I) - (K_2 * I^2)$$

$$C_z = r_z - (K_z * I) - (K_2 * I^2)$$

This model systematically removes both the static bias and the current-induced dynamic interference, and then applies smoothing to produce a stable output that closely represents the true magnetic field B_{true} .

12.7 Model Assumptions

The following assumptions are made:

- Magnetic interference is primarily dependent on motor current
- The relationship between current and interference is approximately linear
- Offset remains constant during operation
- Noise is random and can be reduced using filtering

12.8 Model Limitations

- The linear model may not capture all non-linear interference effects
- Accuracy depends on proper estimation of compensation coefficients
- External magnetic disturbances are not considered
- Performance may vary under different operating conditions

13. RESULTS AND DISCUSSION

13.1 Real Drone Magnetometer Analysis

The real-time magnetometer data obtained from the drone was analyzed to study the behavior of the magnetic field under practical operating conditions. Data was collected during both idle and active motor states using the onboard sensor system.

Magnetic Field Interference in X, Y, Z Directions

The magnetic field variation across all three axes is shown in Figure 13.1. During idle conditions, the magnetometer readings remain relatively stable, reflecting the ambient magnetic field with only minor sensor disturbance. However, when the motor and Electronic Speed Controller (ESC) are activated, significant fluctuations are introduced across all three axes simultaneously.



Figure 13.1 Raw Magnetic Field Strength

These fluctuations are primarily caused by electromagnetic interference generated by the flow of current through the motor windings and associated circuitry. The magnitude of disturbance increases with motor activity, confirming a direct relationship between motor current and magnetic interference. Notably, the interference does not affect all axes equally, as the physical orientation of the motor relative to the sensor introduces axis-dependent variation in the disturbance magnitude.

Thrust vs Magnetic Field

Figure 13.2 presents the relationship between motor thrust and the resulting magnetic field disturbance. A clear trend is observed — as thrust increases, the magnetic field deviation from the baseline also increases. This is consistent with the underlying assumption of the proposed compensation model, where magnetic interference is treated as a function of motor current, which in turn scales with thrust output.

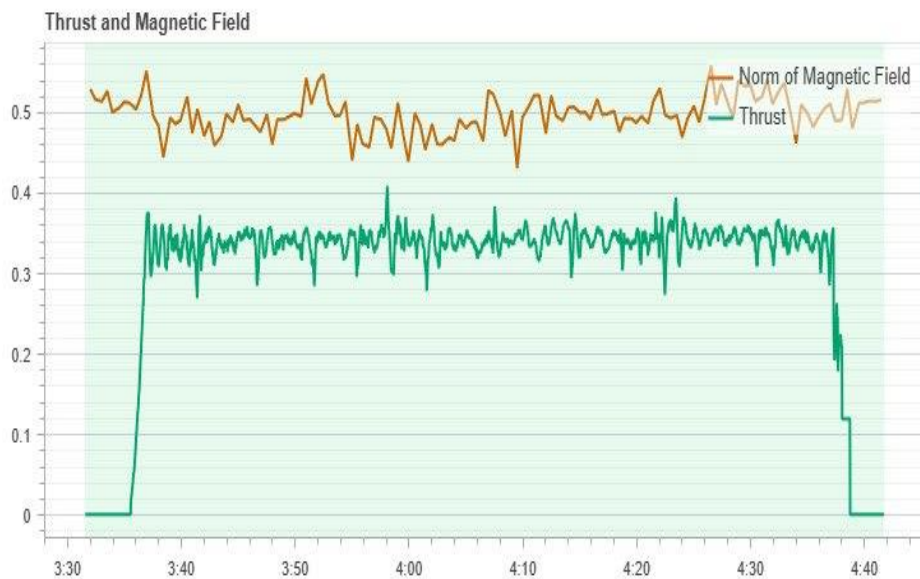


Figure 13.2 Thrust Vs Magnetic Field

This real-time observation confirms that magnetometer measurements in drone systems are significantly affected by motor-induced electromagnetic interference. If uncompensated, disturbances can degrade the accuracy of heading estimation, navigation, and orientation systems. This motivates the compensation approach developed in this work and validates the experimental methodology used to characterize the interference.

13.2 Problem Identification

From the real-time analysis presented in the previous section, it is evident that the magnetometer measurements are affected by multiple sources of error, with motor-induced electromagnetic interference being the most significant contributor.

The key issues identified are:

- **Baseline Offset** — A constant bias is present in the magnetometer readings even under idle conditions, independent of any motor activity.
- **Motor-Induced Fluctuations** — Significant fluctuations are observed in the magnetic field during motor operation, increasing in magnitude with motor speed and current draw.
- **Current-Dependent Interference** — The magnetic interference exhibits a direct relationship with motor current variations, confirming its electromagnetic nature.
- **Sensor interference** — Transient interference and high-frequency disturbances are present in the sensor output, further degrading the quality of the measured signal.

These issues collectively result in inaccurate and unstable magnetic field measurements, which can negatively impact heading estimation and orientation accuracy in drone navigation and control applications.

To address these problems, the proposed work implements a three-stage compensation pipeline:

- **Stage 1** — Baseline offset removal to eliminate the static bias component.
- **Stage 2** — Current-dependent interference estimation and subtraction.
- **Stage 3** — Signal smoothing using an filter (smoothing factor) to suppress residual disturbance.

The implementation and results of this approach are discussed in the following sections.

13.3 Experimental Trails And Observations

13.3.1 Trial 1: Initial Implementation – Nanganallur Location

The results of Trial conducted in **Nanganallur** are presented in **Figure 13.3** and **Figure 13.4**, showing raw and compensated magnetic field data respectively for all three axes.

It can be observed that the compensated signal does not show any significant improvement over the raw signal. The fluctuations in the magnetic field remain largely unchanged, particularly during motor operation, suggesting that the applied compensation had little practical effect.

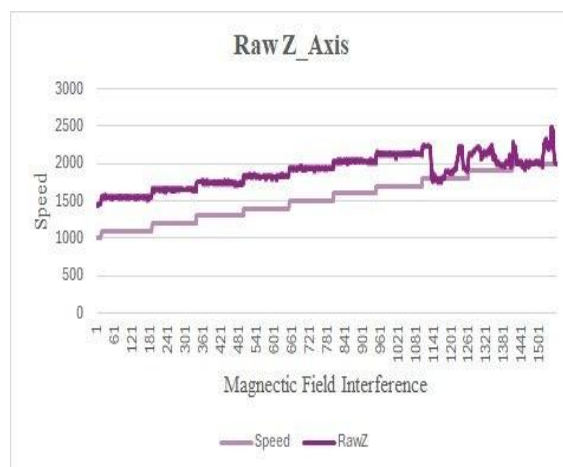
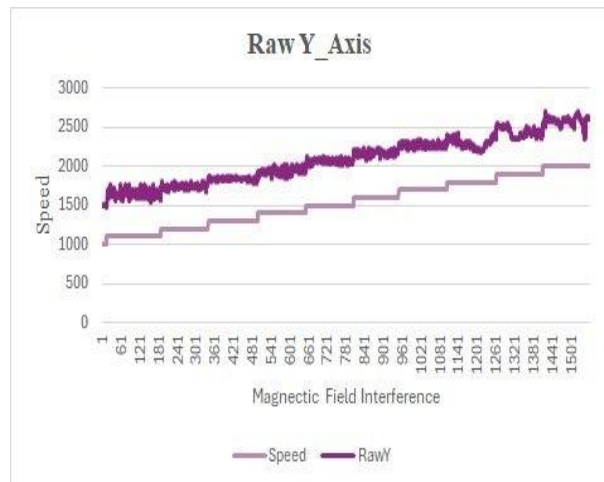
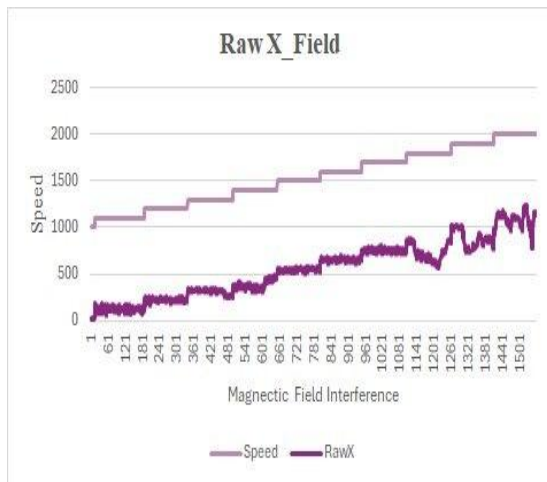
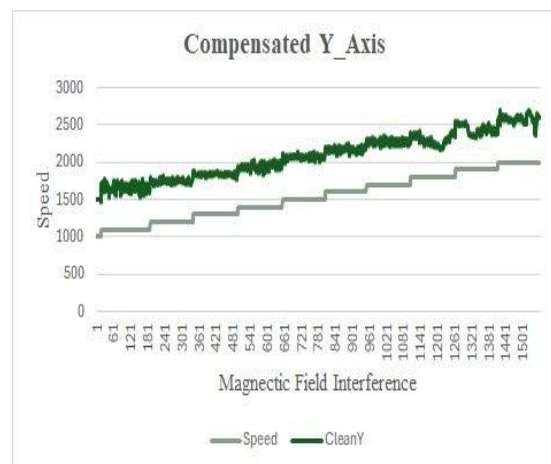


Figure 13.3 Raw Magnetic Field (X, Y, Z)



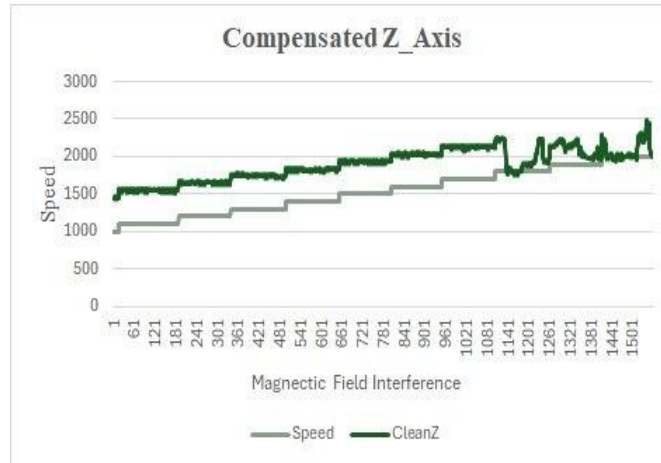


Figure 13.4 Compensated Magnetic Field (X, Y, Z)

Although a current-dependent compensation model was applied, the absence of baseline offset removal rendered the interference suppression ineffective. Since the static bias component was not eliminated prior to compensation, the model operated on already-shifted data, limiting its ability to isolate and remove the dynamic interference. As a result, the compensated signal closely follows the raw signal across all three axes, confirming that the model failed to adequately eliminate both the bias and the motor-induced disturbances.

This outcome is consistent with the limitations identified in the initial model discussed, and highlights that direct current-based compensation without prior offset removal is insufficient for accurate magnetic field correction. These findings motivated the inclusion of baseline offset removal as a preprocessing step in the improved model.

13.3.2 Trial 2: Improved Model – Veppambaattu Location (Partial Compensation)

The results of Trial 2, conducted in **Veppambaattu**, are shown in Figure 13.5, 13.6, 13.7. The graph presents a comparison between the raw and compensated magnetic field values across all three axes (X, Y, Z).

In this trial, baseline offset removal was introduced along with linear compensation. The compensated signals show noticeable improvement compared to the raw data, with reduced baseline shift and partial stabilization.

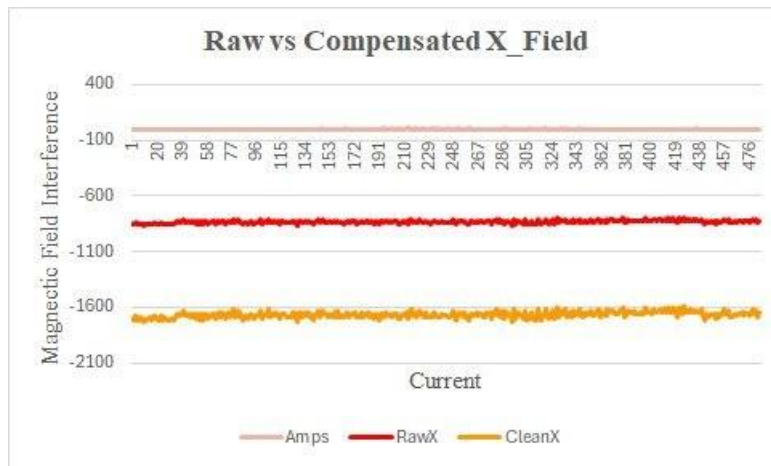


Figure 13.5 Raw Vs Compensated Magnetic Field (X - Direction)

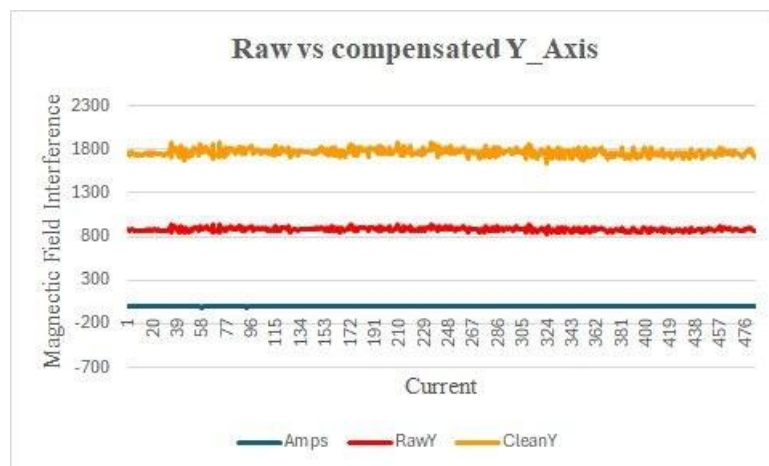


Figure 13.6 Raw Vs Compensated Magnetic Field (Y - Direction)

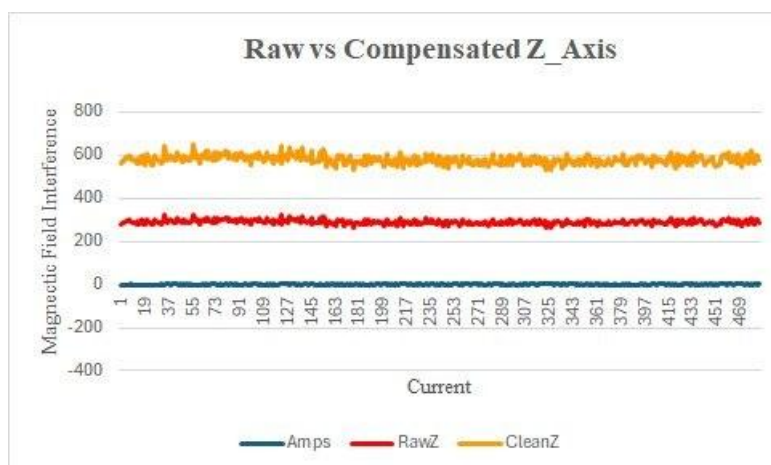


Figure 13.7 Raw Vs Compensated Magnetic Field (Z - Direction)

However, fluctuations are still present in the signal, particularly during variations in motor speed. This indicates that although offset removal improves accuracy, the compensation is not yet fully effective.

The absence of a fully stable output suggests that further refinement, including proper tuning of compensation coefficients and filtering, is required

13.3.3 Trial 3: Final Compensated Model – Veppampattu Location

The results of the final trial, conducted in Veppampattu, are presented in Figure 13.8,13.9,13.10, which compares the raw and compensated magnetic field values across all three axes (X, Y, Z).

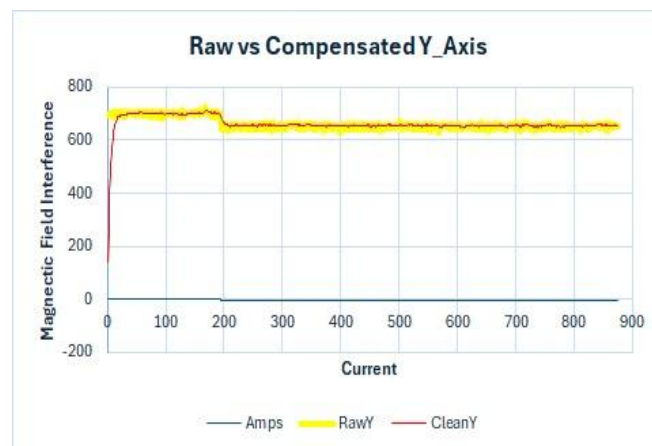


Figure 13.8 Raw Vs Compensated Magnetic Field (X - Direction)

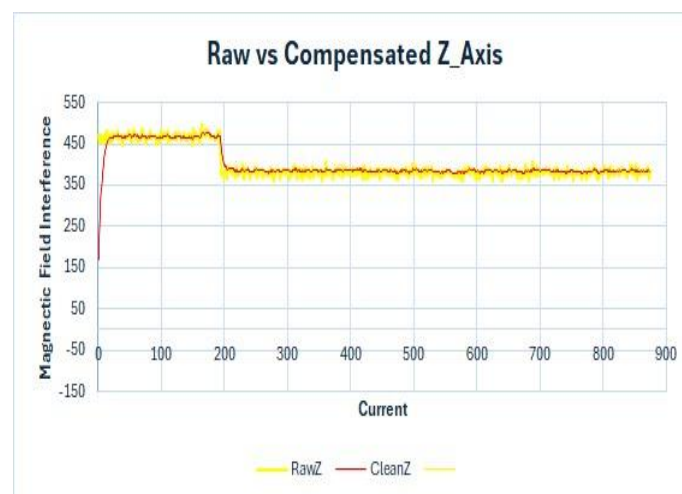


Figure 13.9 Raw Vs Compensated Magnetic Field (Y - Direction)



Figure 13.10 Raw Vs Compensated Magnetic Field (Z - Direction)

In this trial, the complete three-stage compensation pipeline was implemented, incorporating baseline offset removal, properly tuned linear compensation coefficients (K_x , K_y , K_z), and an smoothing filter with a smoothing factor of $\alpha=0.8$.

The results clearly demonstrate a significant improvement in compensation performance compared to both Trial 1 and Trial 2. The compensated signal is stable and exhibits minimal fluctuations across all three axes, even during active motor operation. The motor-induced magnetic interference, which was prominently visible in the raw signal, is effectively suppressed in the compensated output.

The compensated signal remains nearly constant throughout the observation period, confirming the successful removal of both the static baseline offset and the dynamic current-induced interference components. Furthermore, the application of the smoothing filter effectively reduces residual magnetic interference and transient disturbances, contributing to a cleaner and more reliable output signal.

These results validate the effectiveness of the proposed compensation model and confirm that the combination of offset removal, linear compensation, and signal smoothing is capable of producing accurate and stable magnetometer measurements under practical drone operating conditions. The progressive improvement observed across the three trials — from Trial 1 through to the final implementation — demonstrates the importance of each processing stage in achieving the overall compensation objective.

13.4 Comparative Analysis

A comparative analysis was performed to evaluate the performance of the compensation approach across all three experimental trials. The comparison focuses on the effectiveness of each model in reducing magnetic field fluctuations and improving signal stability.

In Trial 1, conducted in Nanganallur, compensation was applied without baseline offset removal. As a result, the compensated signal showed minimal improvement over the raw data, with significant fluctuations remaining throughout motor operation, indicating ineffective compensation.

In Trial 2, conducted in Veppampattu, baseline offset removal was introduced alongside linear compensation. This resulted in partial improvement, where the signal showed reduced bias and moderate stabilization. However, noticeable fluctuations were still present, particularly during variations in motor speed.

In Trial 3, also conducted in Veppampattu, the complete compensation model was implemented incorporating baseline offset removal, properly tuned linear compensation coefficients, and an EMA smoothing filter. This produced a stable and consistent output signal with significantly reduced fluctuations, confirming effective suppression of both static and dynamic magnetic interference.

Table 13.1 Comparative Summary of Trials

Trail	Location	Method used	Result
Trail 1	Nanganallur	Linear compensation only (no offset removal)	No effective compensation
Trail 2	Veppampattu	Offset removal + Linear compensation	Partial improvement
Trail 3	Veppampattu	Offset removal + Linear compensation + smoothing Filter	Effective compensation

The progressive improvement observed across the three trials clearly demonstrates that the combination of offset removal, linear compensation, and smoothing filtering provides the best compensation performance. Each stage contributes meaningfully to the overall result, and no single step alone is sufficient for accurate magnetic field correction.

13.5 Performance Evaluation

The performance of the proposed compensation method was quantitatively evaluated by comparing the variation in raw and compensated magnetic field data.

Two key metrics were used for evaluation: range reduction and standard deviation reduction.

Range-based evaluation:

$$\text{Percentage Reduction (\%)} = [(\text{Range_raw} - \text{Range_comp}) / \text{Range_raw}] \times 100$$

Standard deviation-based evaluation:

$$\text{Percentage Reduction (\%)} = [(\sigma_{\text{raw}} - \sigma_{\text{comp}}) / \sigma_{\text{raw}}] \times 100$$

Table 13.2 Performance Results

Axis	Reduction (%)
X-Axis	26%
Y-Axis	46%
Z-Axis	27%
Overall	54%

The results indicate that the proposed compensation method significantly reduces magnetic field fluctuations. The highest improvement is observed along the Y-axis, while the X and Z axes also show noticeable reduction in variation.

The overall improvement of approximately 54% confirms that the model effectively minimizes current-induced magnetic interference and enhances signal stability.

13.6 Discussion

The results obtained from the experimental trials demonstrate the effectiveness of the proposed compensation model in reducing magnetic field interference.

In the first trial, the absence of offset removal resulted in poor compensation performance. The presence of constant bias in the data prevented accurate interference estimation, leading to unstable output.

The second trial showed improvement after introducing baseline offset removal. Although the signal became more stable, fluctuations were still present due to incomplete compensation and lack of filtering.

In the final trial, the complete model was implemented, including offset removal, properly tuned linear compensation, and smoothing filtering. This resulted in a stable and consistent signal with significantly reduced fluctuations.

The performance evaluation results further validate the effectiveness of the proposed method, achieving an overall improvement of approximately 54%. This confirms that the linear compensation model is sufficient for the given system, and the contribution of non-linear effects is minimal.

Additionally, the smoothing filter played a crucial role in reducing noise and improving signal quality.

Overall, the proposed approach provides a simple and efficient solution for magnetic field compensation, making it suitable for real-time embedded applications.

14. CONCLUSION

14.1 Conclusion

This project successfully developed and implemented a magnetic field compensation system for reducing motor-induced electromagnetic interference in magnetometer measurements. In motor-driven systems such as drones and embedded platforms, the operation of the motor and Electronic Speed Controller (ESC) introduces unwanted magnetic disturbances that affect the accuracy and stability of sensor readings. The primary objective of this work was to identify these disturbances and develop a practical compensation method capable of improving measurement reliability in real-time conditions.

To address this problem, a systematic methodology was adopted involving hardware integration, sensor calibration, real-time data acquisition, mathematical modeling, and signal processing. A magnetometer sensor was used to measure the surrounding magnetic field, while a current sensor monitored the motor current responsible for interference generation. Experimental observations confirmed that variations in motor current directly influenced the magnetometer output, resulting in fluctuations and instability in the measured data.

Based on this relationship, a current-dependent compensation model was developed. Initially, baseline offset removal was implemented to eliminate constant bias present in the raw sensor data. After offset correction, a linear compensation method was applied to estimate and subtract the interference component proportional to motor current. In addition, a smoothing filter based on the Exponential Moving Average (EMA) technique was introduced to suppress high-frequency noise and transient fluctuations.

Three experimental trials were conducted to validate the proposed approach. The first trial, performed without offset correction, showed poor compensation performance and significant signal variation. The second trial introduced offset removal and linear compensation, resulting in partial improvement. The final trial implemented the complete model with coefficient tuning and filtering, producing stable and consistent magnetic field measurements with significantly reduced fluctuations.

Performance evaluation was carried out using range reduction and standard deviation analysis. The final compensated output demonstrated an overall improvement of approximately 54% in magnetic field stability compared to the raw signal. These results

confirm that the proposed compensation method effectively minimizes motor-induced interference and enhances sensor reliability.

The project demonstrates that a simple current-based linear compensation model, when combined with proper offset removal and filtering, can provide an efficient and practical solution for real-time magnetic field correction. The proposed system offers low computational complexity, ease of implementation, and suitability for embedded platforms where processing resources are limited.

In conclusion, the developed magnetic field compensation system achieved the intended objectives and provides a valuable approach for improving magnetometer performance in drones, robotic systems, and other motor-driven applications requiring accurate orientation and navigation sensing.

14.2 Future Scope

The present work can be further extended in several ways to improve system performance and applicability:

- Implementation of adaptive compensation techniques for changing operating conditions.
- Use of machine learning methods for more accurate interference prediction.
- Integration with GPS and IMU sensors for advanced drone navigation systems.
- Development of wireless real-time monitoring interfaces.
- Testing under different environmental and industrial operating conditions.
- Miniaturization of the system for compact UAV and portable embedded applications.

Thus, the proposed work creates a strong foundation for future research in intelligent magnetic field compensation systems.

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